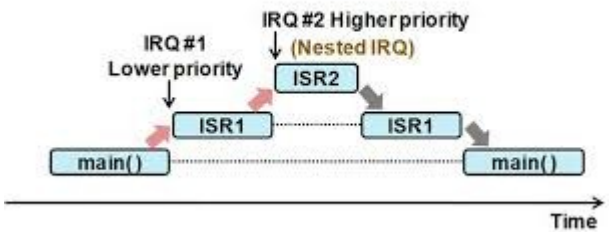
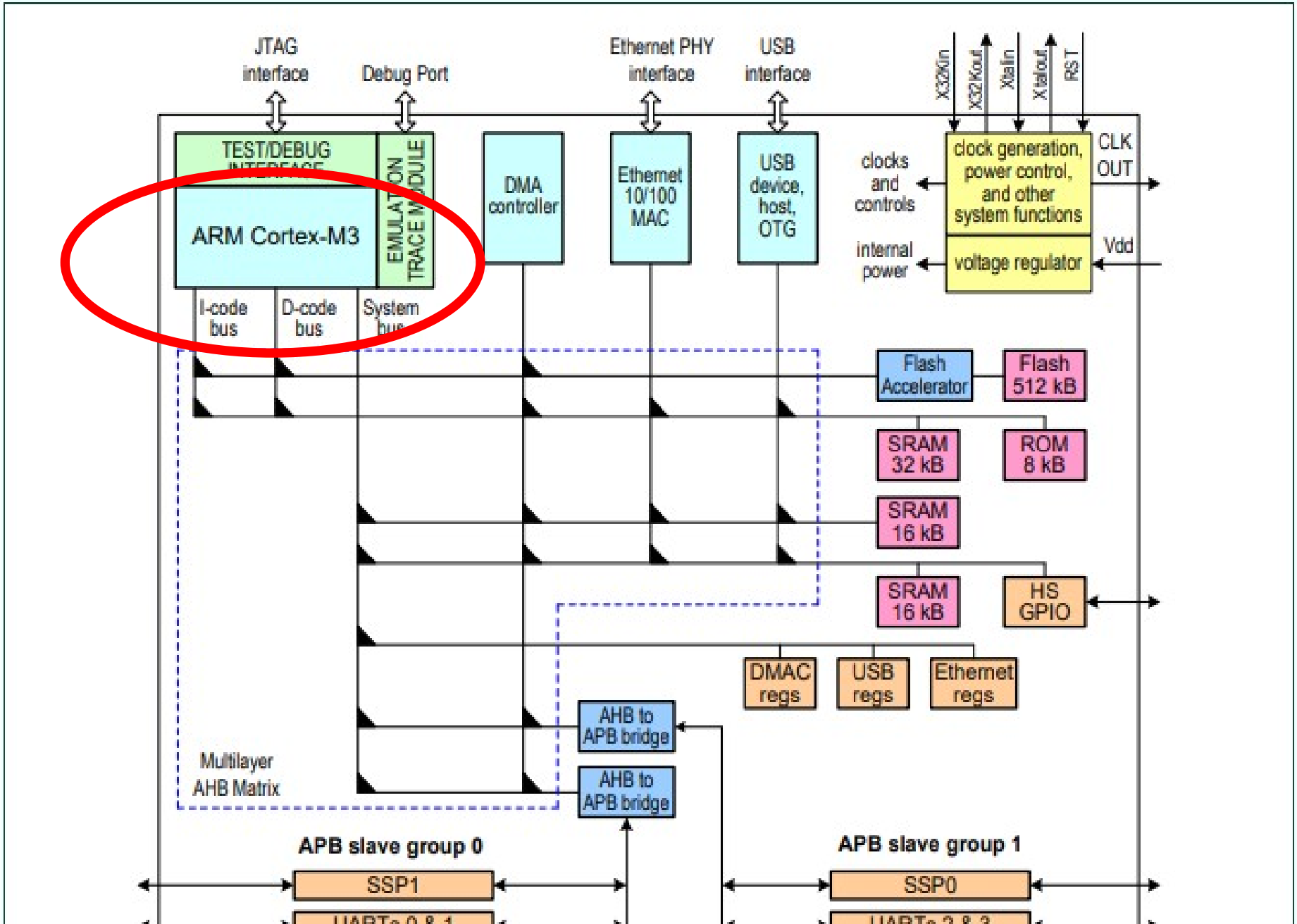
ELECTRONICA DIGITAL III

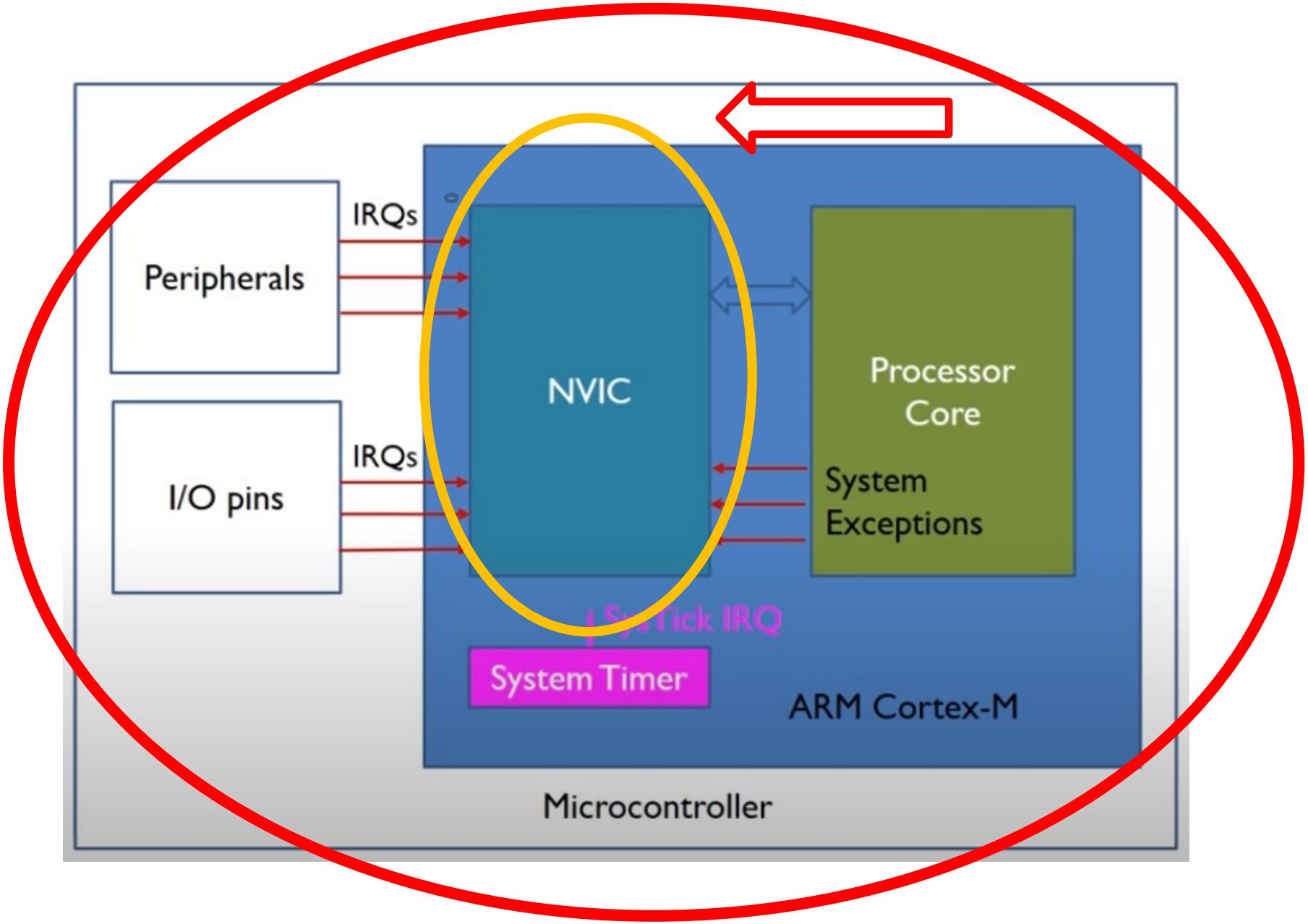
ARM - Cortex M3
Advanced RISC Machines

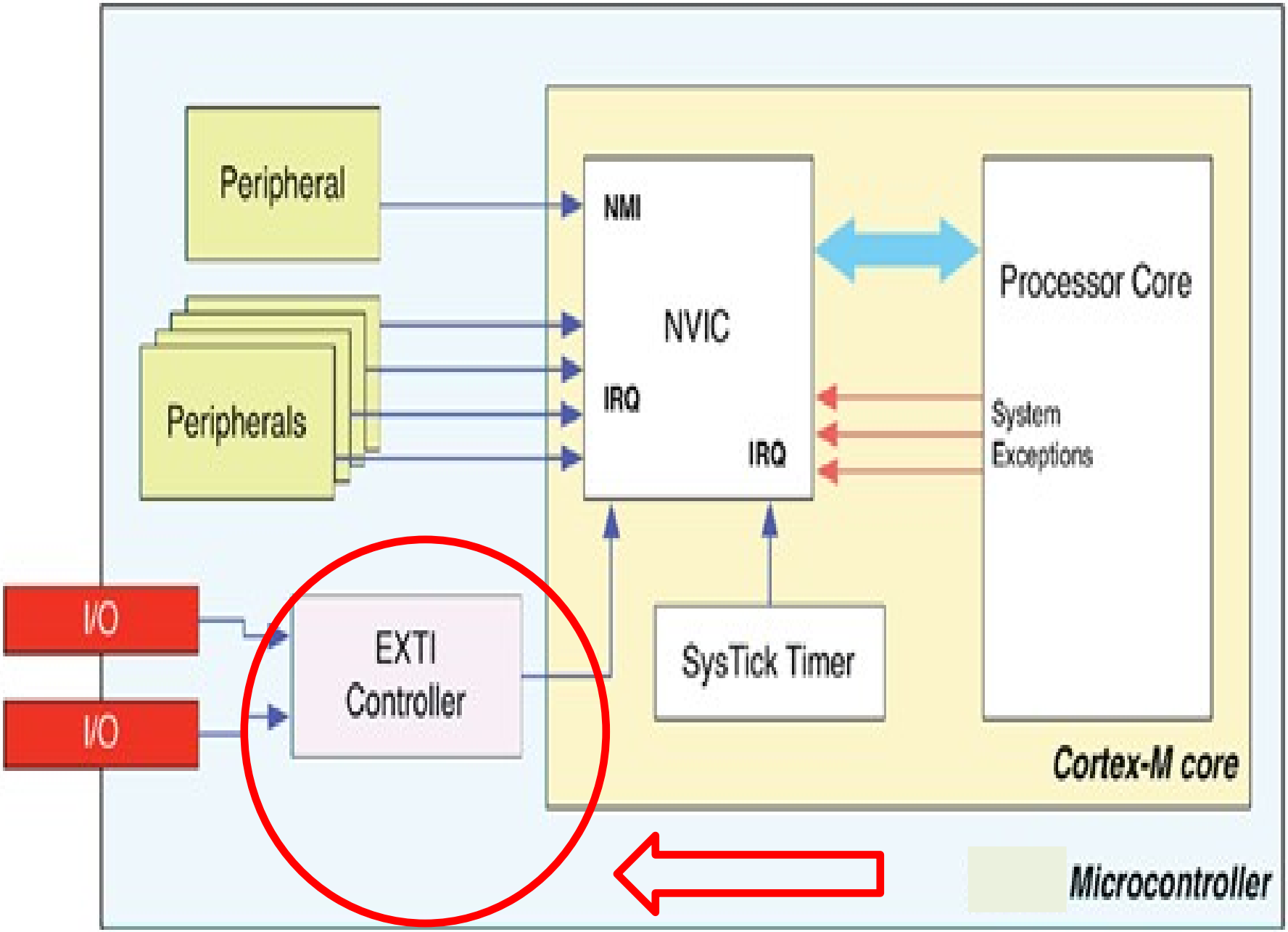
INTERRUPCIONES NVIC



1.10 Block diagram




Peripherals
IRQs
I/O pins
IRQs
NVIC
Processor
Core
System
Exceptions
SysTick IRQ
System Timer
ARM Cortex-M
Microcontroller


Peripheral
Peripherals
NMI
NVIC
IRQ
IRQ
Processor Core
System
Exceptions
I/O
I/O
EXTI
Controller
SysTick Timer
Cortex-M core
Microcontroller

Nested Vectored Controlador de Interrupciones (NVIC)

Controlador de interrupción vectorial anidado

- El controlador de interrupciones proporciona una latencia de interrupción baja.

- Controla las excepciones del sistema y las interrupciones de los periféricos.
- En el LPC17xx, el NVIC admite 35 interrupciones vectoriales.

- 32 niveles de prioridad de interrupción programables, con enmascaramiento de nivel de prioridad de hardware.

- Tabla de vectores reubicable.

- Interrupción no enmascarable.

EXCEPCIONES

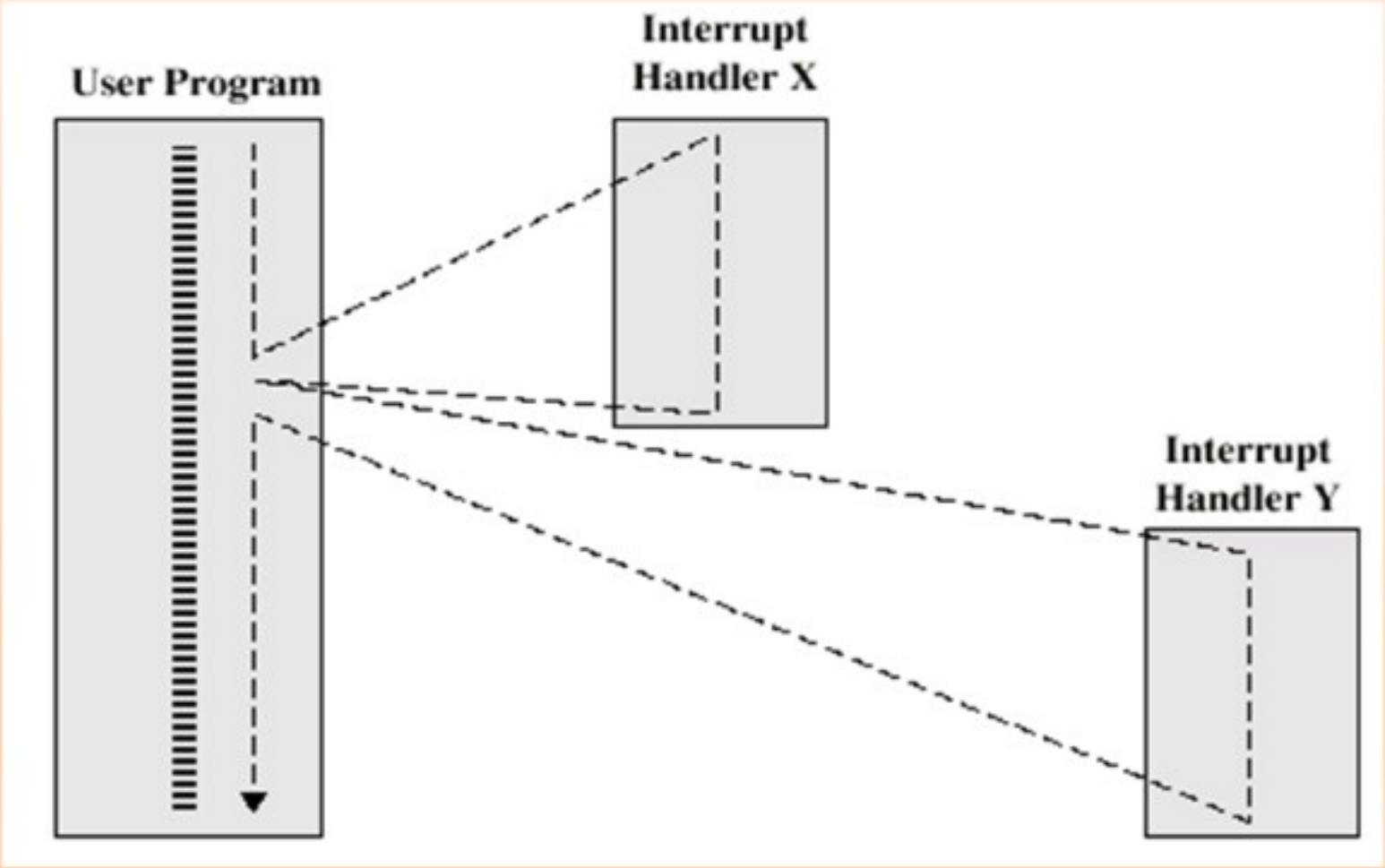
SISTEMA

INTERRUPCIONES

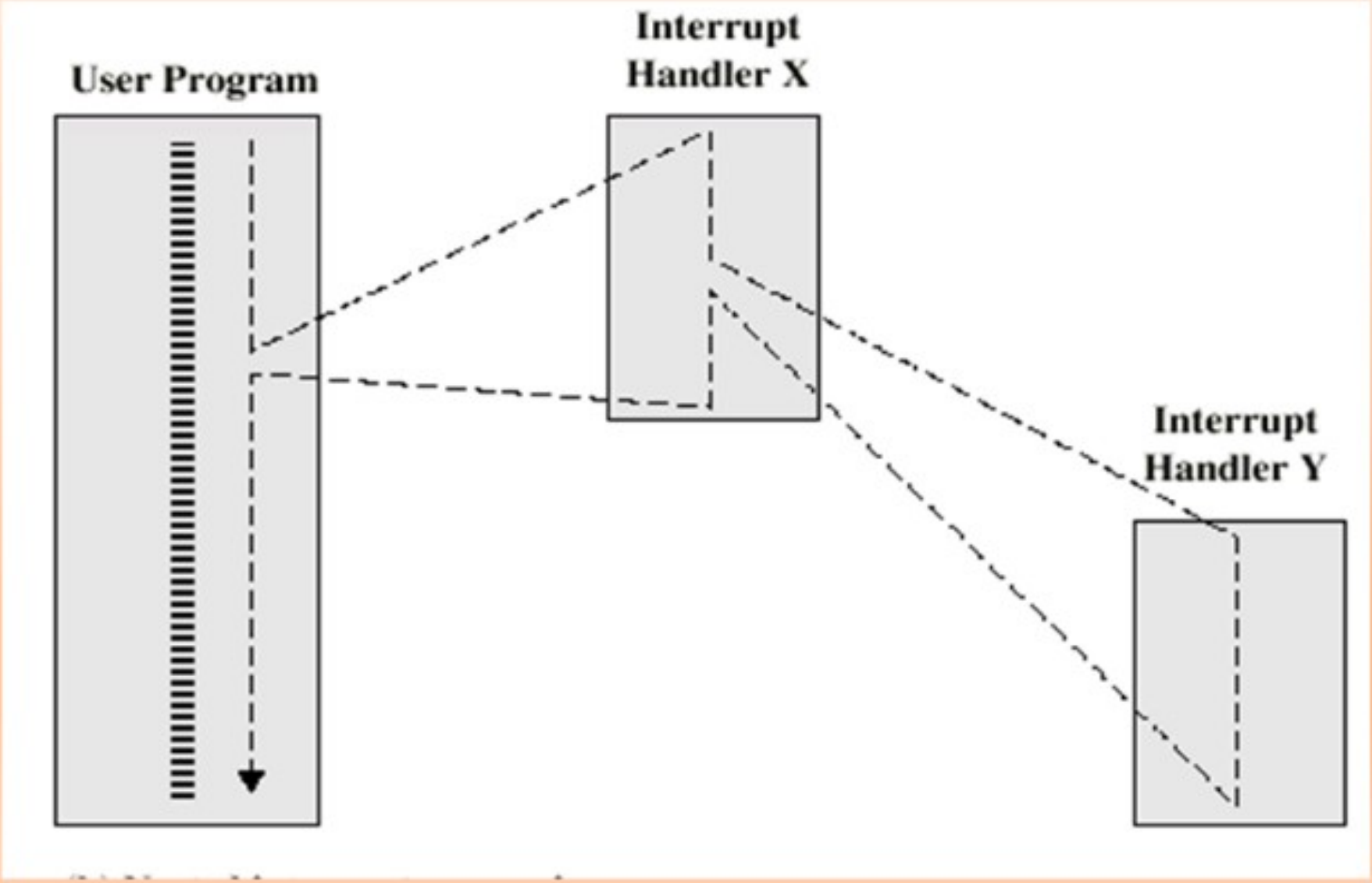
PERIFERICOS

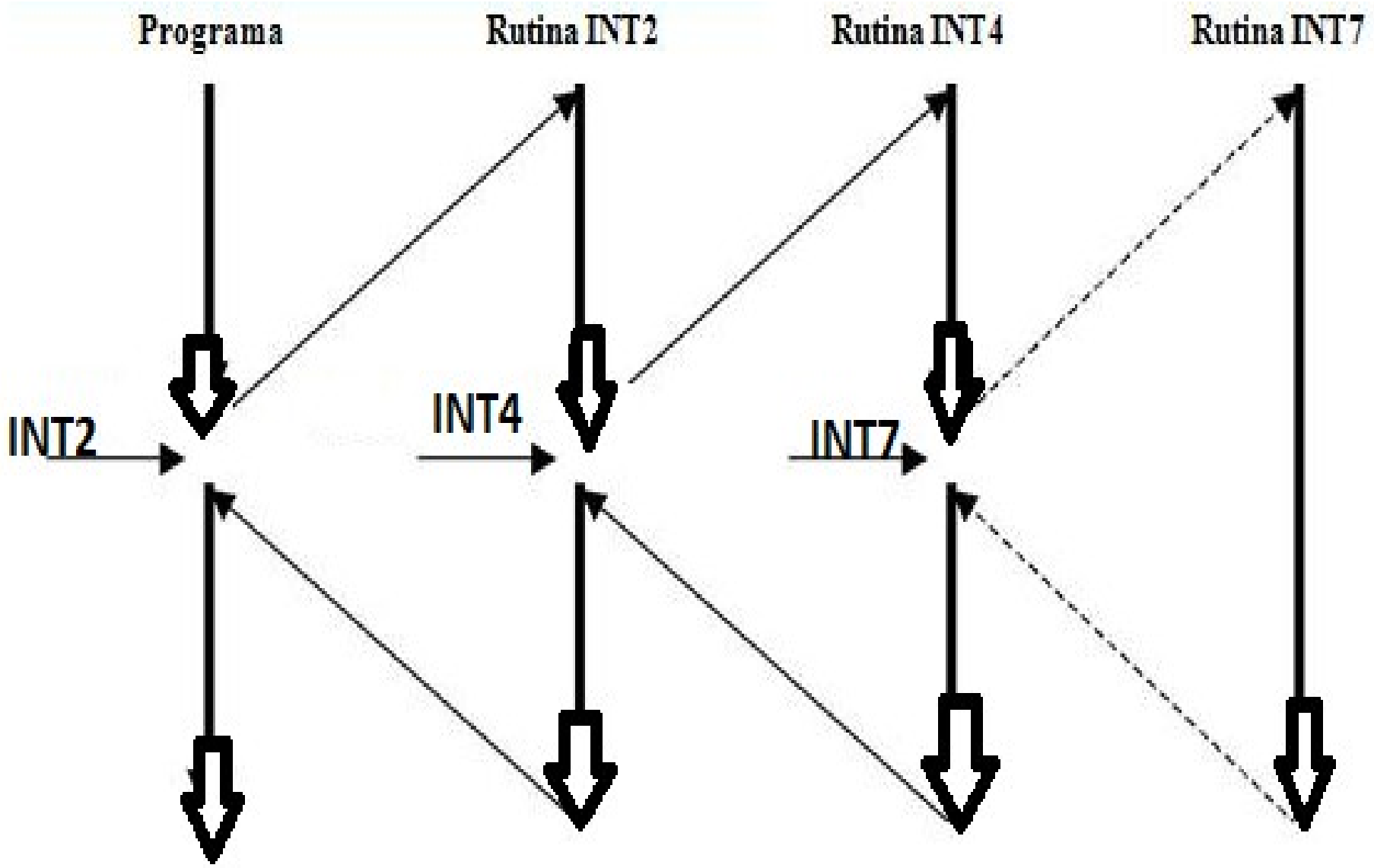
ANIDADAS

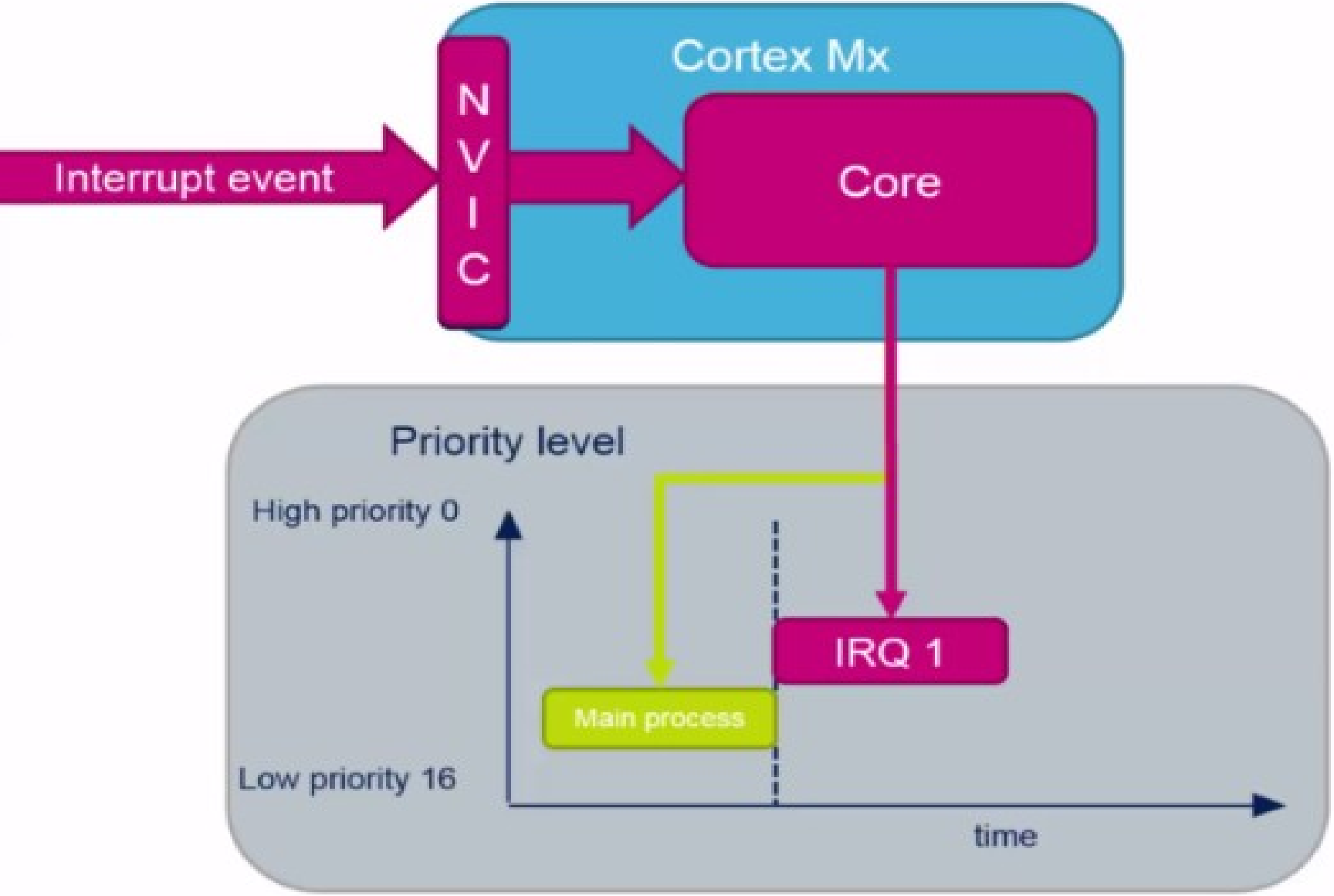
Múltiples Interrupciones - Secuencial



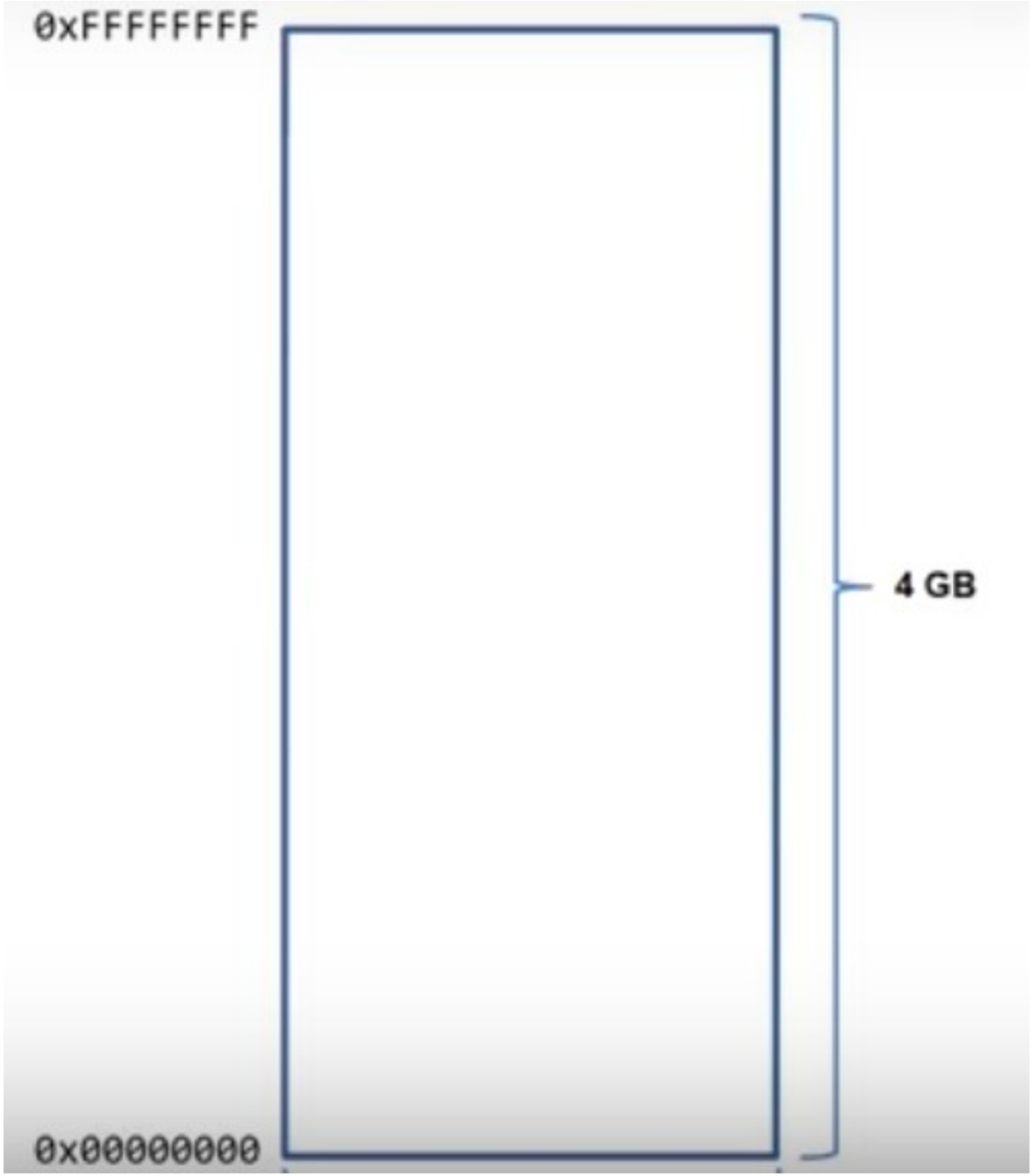
Múltiples Interrupciones – Anidadas

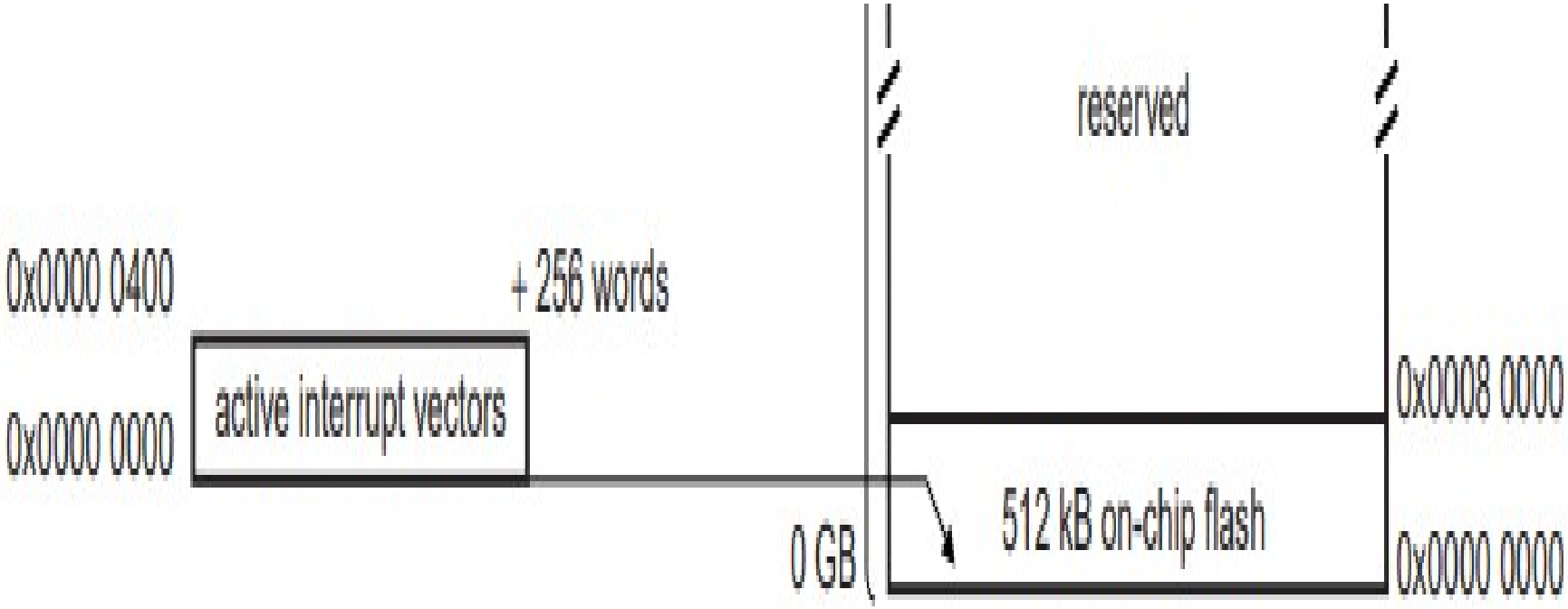


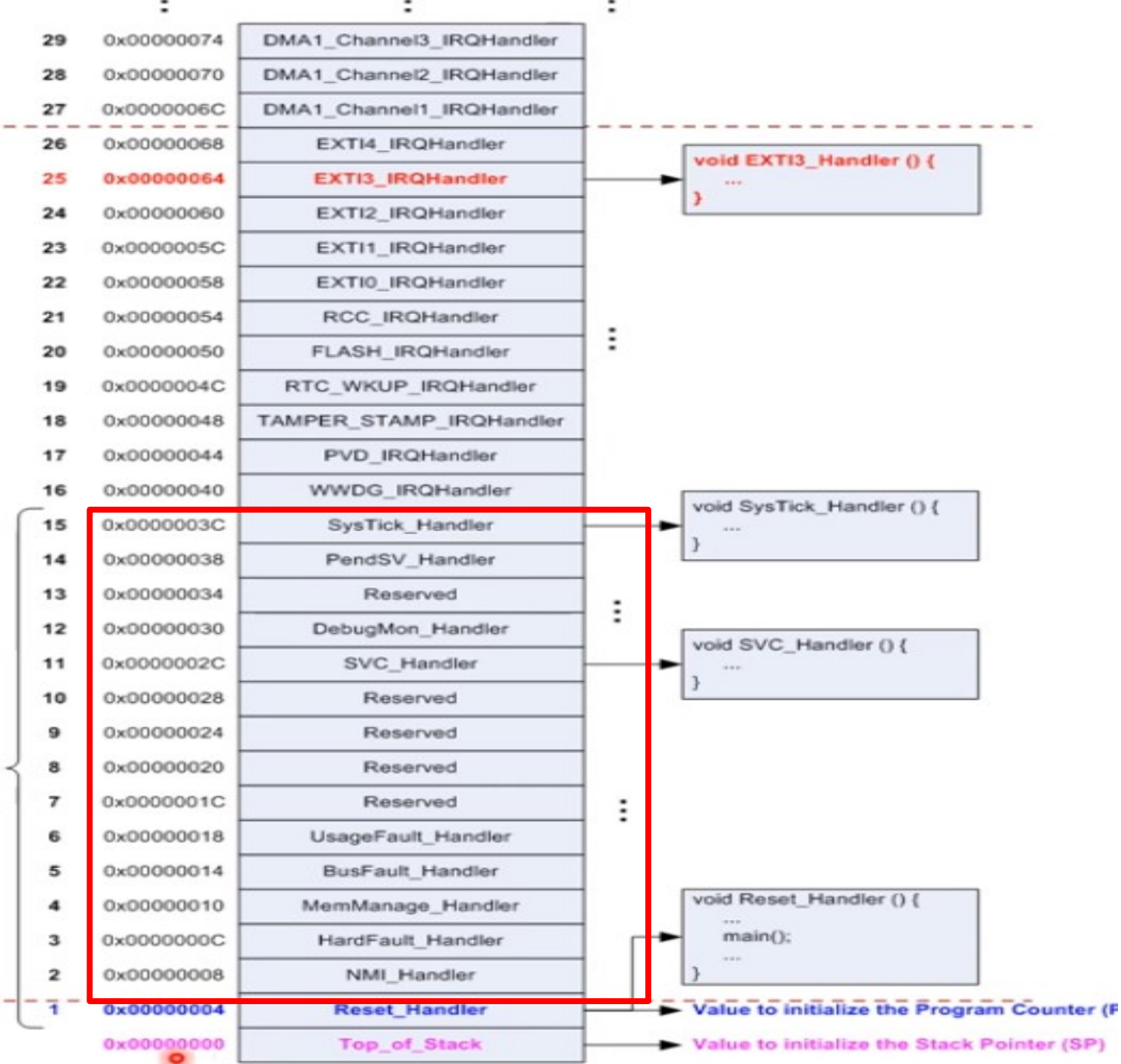

Programa
Rutina INT2
Rutina INT4
Rutina INT7
INT2
INT4
INT7

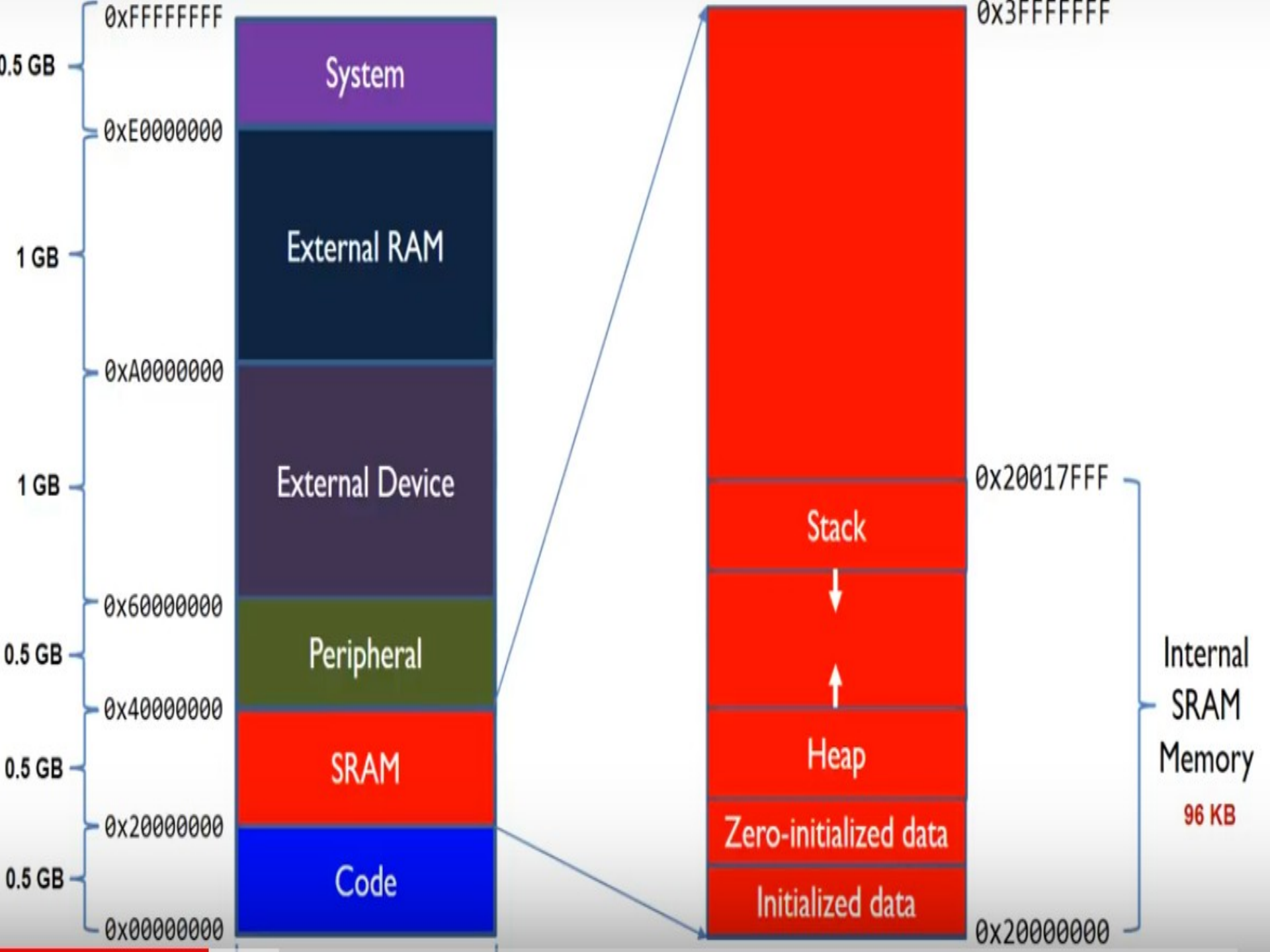

Cortex Mx
NVIC
Interrupt event
Core
Priority level
High priority 0
IRQ 1
Main process
Low priority 16
time

VECTORIZADAS

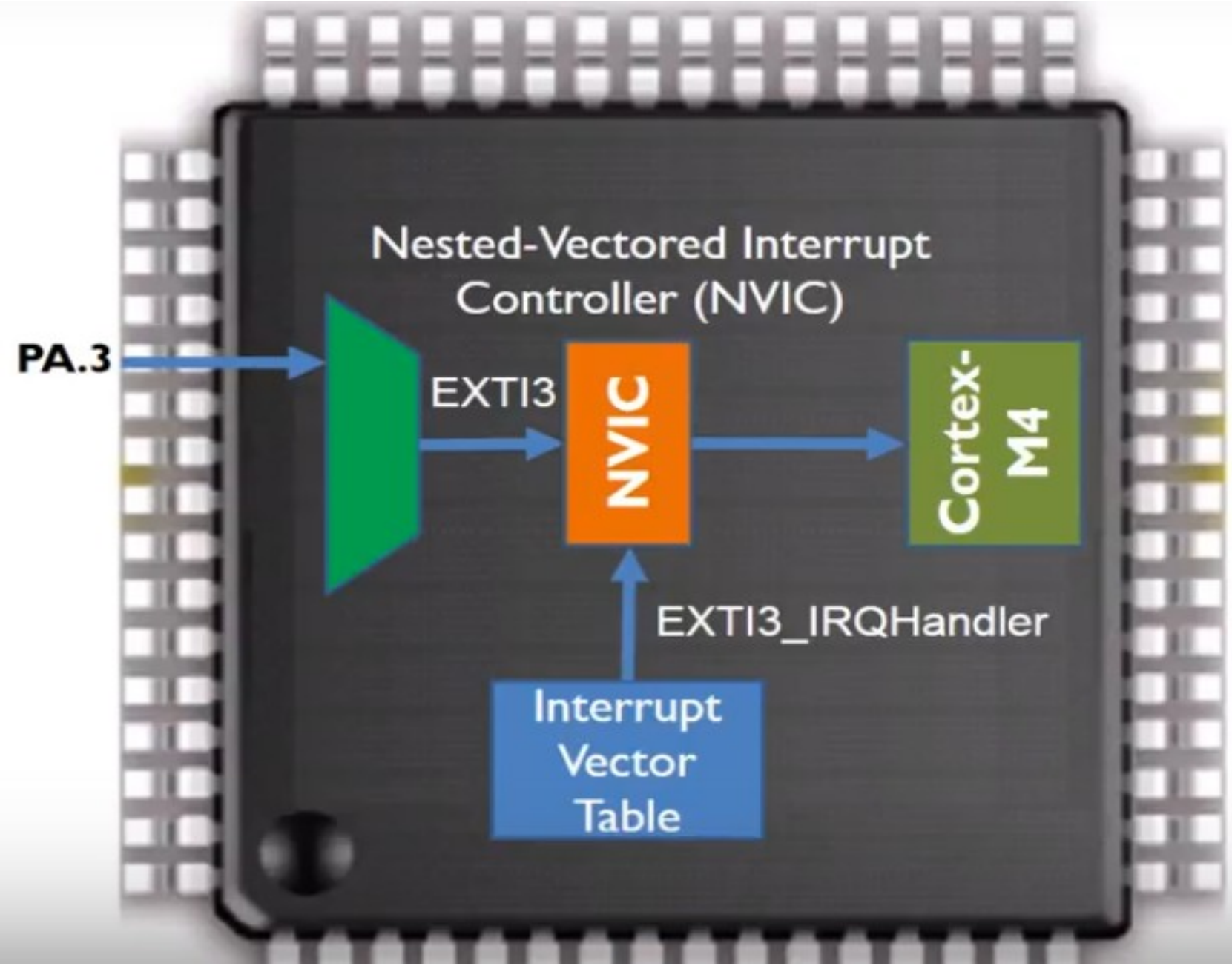

0xFFFFFFFF
4 GB
0x00000000


reserved
0x0000 0400
+ 256 words
active interrupt vectors
0x0000 0000
0x0008 0000
512 kB on-chip flash
0 GB
0x0000 0000


29 0x00000074 DMA1_Channel3_IRQHandler
28 0x00000070 DMA1_Channel2_IRQHandler
27 0x0000006C DMA1_Channel1_IRQHandler
26 0x00000068 EXTI4_IRQHandler
25 0x00000064 EXTI3_IRQHandler
24 0x00000060 EXTI2_IRQHandler
23 0x0000005C EXTI1_IRQHandler
22 0x00000058 EXTI0_IRQHandler
21 0x00000054 RCC_IRQHandler
20 0x00000050 FLASH_IRQHandler
19 0x0000004C RTC_WKUP_IRQHandler
18 0x00000048 TAMPER_STAMP_IRQHandler
17 0x00000044 PVD_IRQHandler
16 0x00000040 WWDG_IRQHandler
15 0x0000003C SysTick_Handler
14 0x00000038 PendSV_Handler
13 0x00000034 Reserved
12 0x00000030 DebugMon_Handler
11 0x0000002C SVC_Handler
10 0x00000028 Reserved
9 0x00000024 Reserved
8 0x00000020 Reserved
7 0x0000001C Reserved
6 0x00000018 UsageFault_Handler
5 0x00000014 BusFault_Handler
4 0x00000010 MemManage_Handler
3 0x0000000C HardFault_Handler
2 0x00000008 NMI_Handler
1 0x00000004 Reset_Handler
0x00000000 Top_of_Stack
void EXTI3_Handler () {
...
}
void SysTick_Handler () {
...
}
void SVC_Handler () {
...
}
void Reset_Handler () {
...
main();
...
}
Value to initialize the Program Counter (P
Value to initialize the Stack Pointer (SP)


0xFFFFFFFF
0.5 GB
0xE0000000
1 GB
0xA0000000
1 GB
0x60000000
0.5 GB
0x40000000
0.5 GB
0x20000000
0.5 GB
0x00000000
System
External RAM
External Device
Peripheral
SRAM
Code
0x3FFFFFFF
0x20017FFF
Stack
Heap
Zero-initialized data
Initialized data
0x20000000
Internal
SRAM
Memory
96 KB

EXTI3_IRQHandler.


Nested-Vectored Interrupt
Controller (NVIC)
PA.3
EXTI3
NVIC
Cortex-
M4
EXTI3_IRQHandler
Interrupt
Vector
Table

Para comprender las interrupciones, el concepto de tabla de vectores de interrupciones es muy importante.

Interrupt Vector Table

Interrupt Number (8 bits)	Memory Address of ISR (32 bits)
1	Interrupt Service Routine for interrupt 1
2	Interrupt Service Routine for interrupt 2
3	Interrupt Service Routine for interrupt 3
4	Interrupt Service Routine for interrupt 4
5	Interrupt Service Routine for interrupt 5
…	…

When interrupt x is triggered, jump to the ISR for interrupt x. ($1 \leq x \leq 255$)

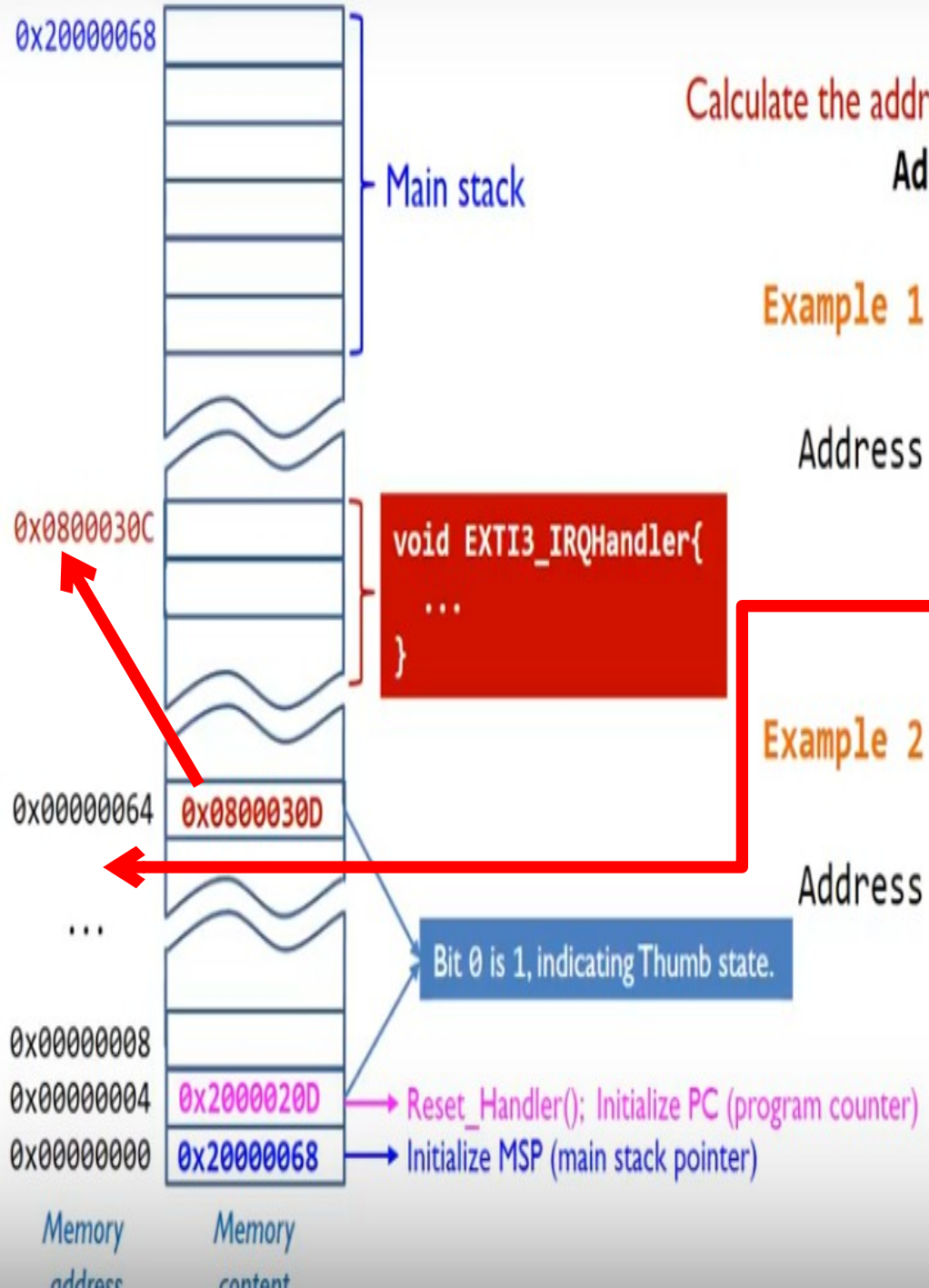
Table 50. Connection of interrupt sources to the Vectored Interrupt Controller

Interrupt ID	Exception Number	Vector Offset	Function	Flag(s)
14	30	0x78	SSP0	Tx FIFO half empty of SSP0 Rx FIFO half full of SSP0 Rx Timeout of SSP0 Rx Overrun of SSP0
15	31	0x7C	SSP 1	Tx FIFO half empty Rx FIFO half full Rx Timeout Rx Overrun
16	32	0x80	PLL0 (Main PLL)	PLL0 Lock (PLOCK0)
17	33	0x84	RTC	Counter Increment (RTCCIF) Alarm (RTCALF)
18	34	0x88	External Interrupt	External Interrupt 0 (EINT0)
19	35	0x8C	External Interrupt	External Interrupt 1 (EINT1)
20	36	0x90	External Interrupt	External Interrupt 2 (EINT2)
21	37	0x94	External Interrupt	External Interrupt 3 (EINT3). **Note:** EINT3 channel is shared with GPIO interrupts
22	38	0x98	ADC	A/D Converter end of conversion
23	39	0x9C	BOD	Brown Out detect

En pocas palabras, la tabla de vectores de interrupción contiene una matriz de punteros de función.

El número de interrupción se utiliza para indexar en la tabla de vectores de interrupción.

¿Cómo utiliza el controlador NVIC el número de interrupción para buscar la tabla de vectores de interrupción?



Calculate the address which holds the address of the ISR for interrupt n:

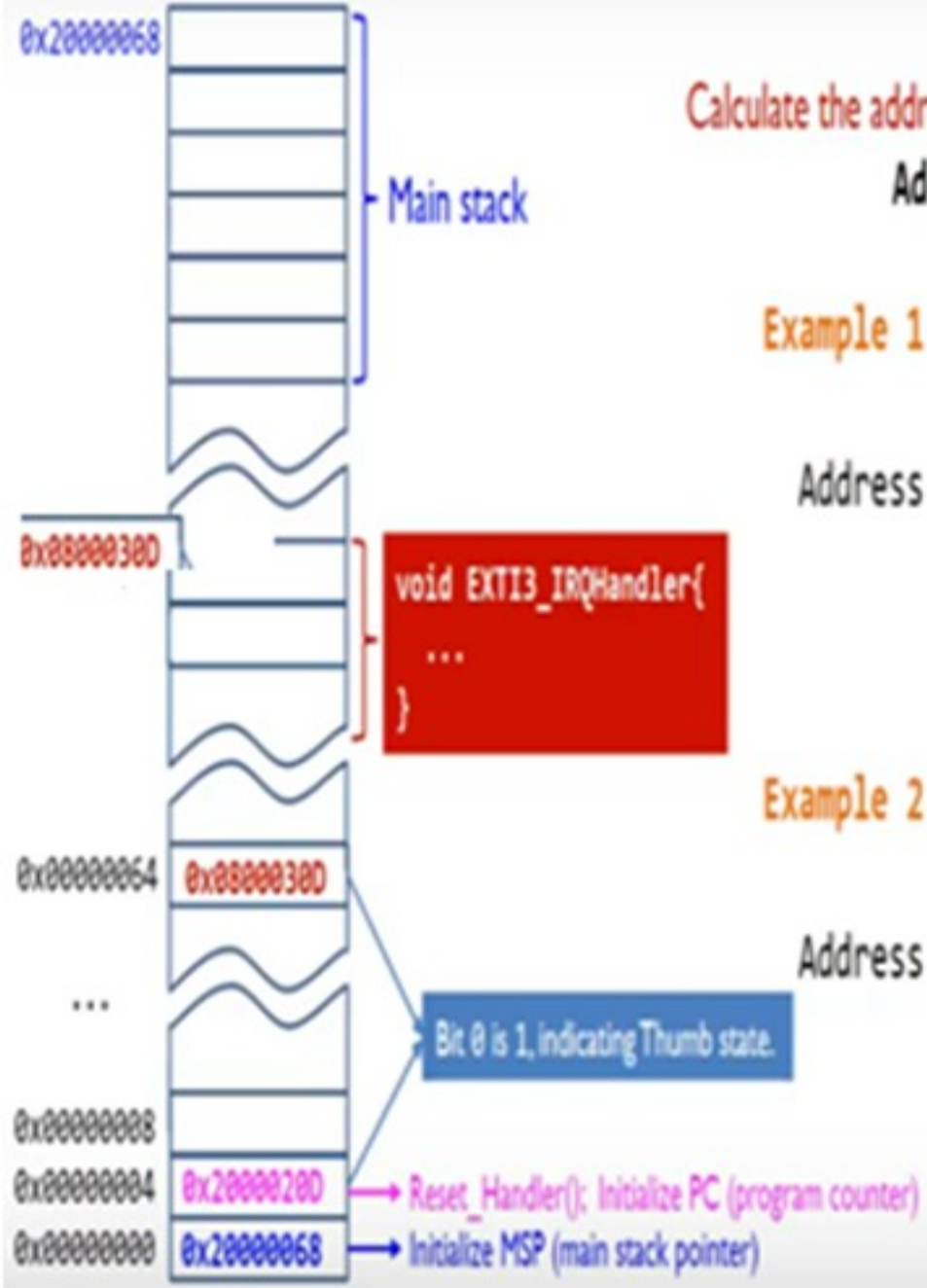
Address of pointer = 64 + 4 × n

Example 1: EXTI3_IRQn = 9

Address of pointer to EXTI3 ISR = 64 + 4 × 9
= 100
= 0x64

Example 2: SysTick_IRQn = -1

Address of pointer to SysTick ISR = 64 + 4 × (-1)
= 60
= 0x3C



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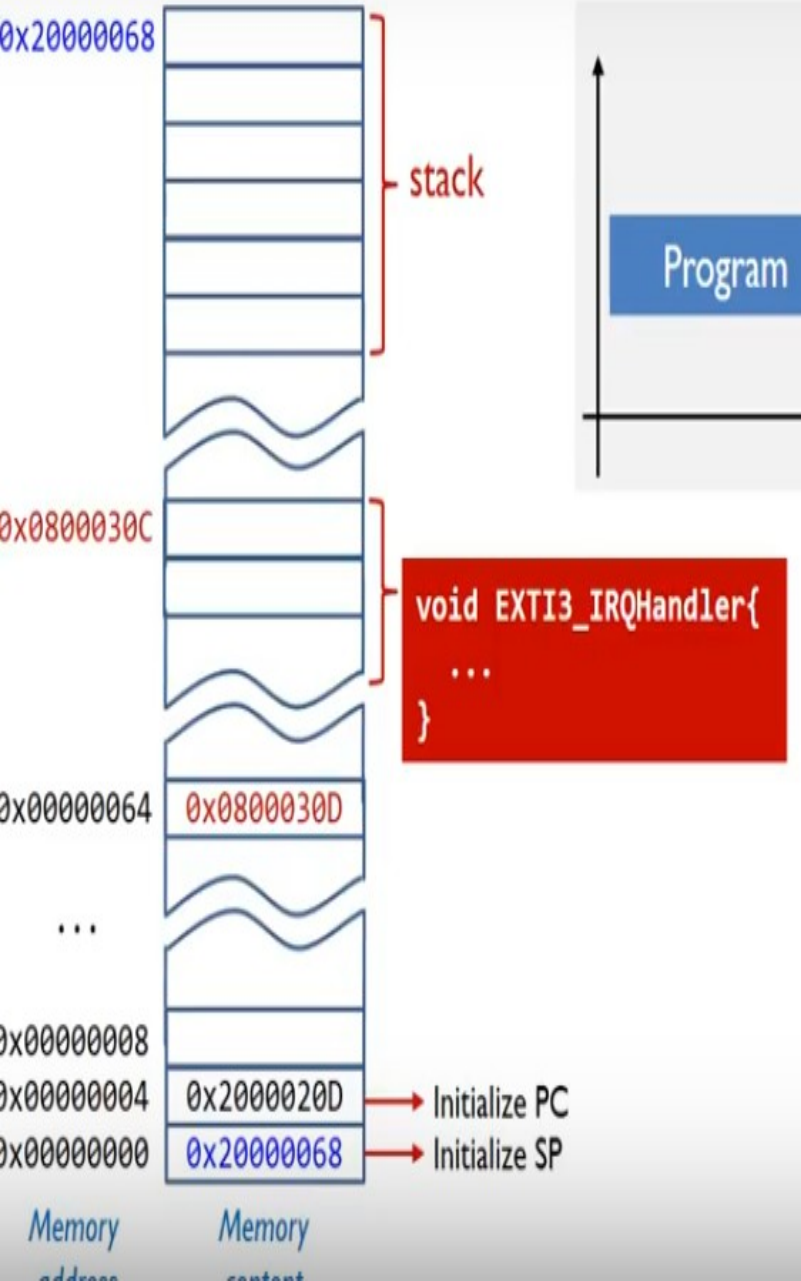
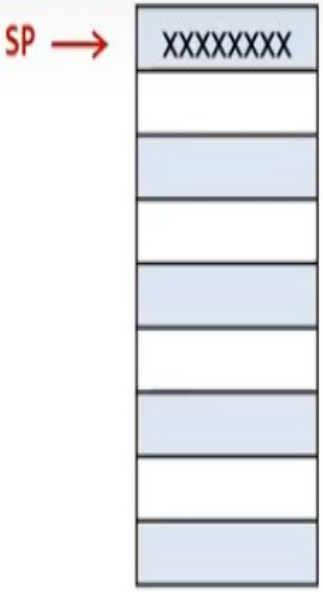
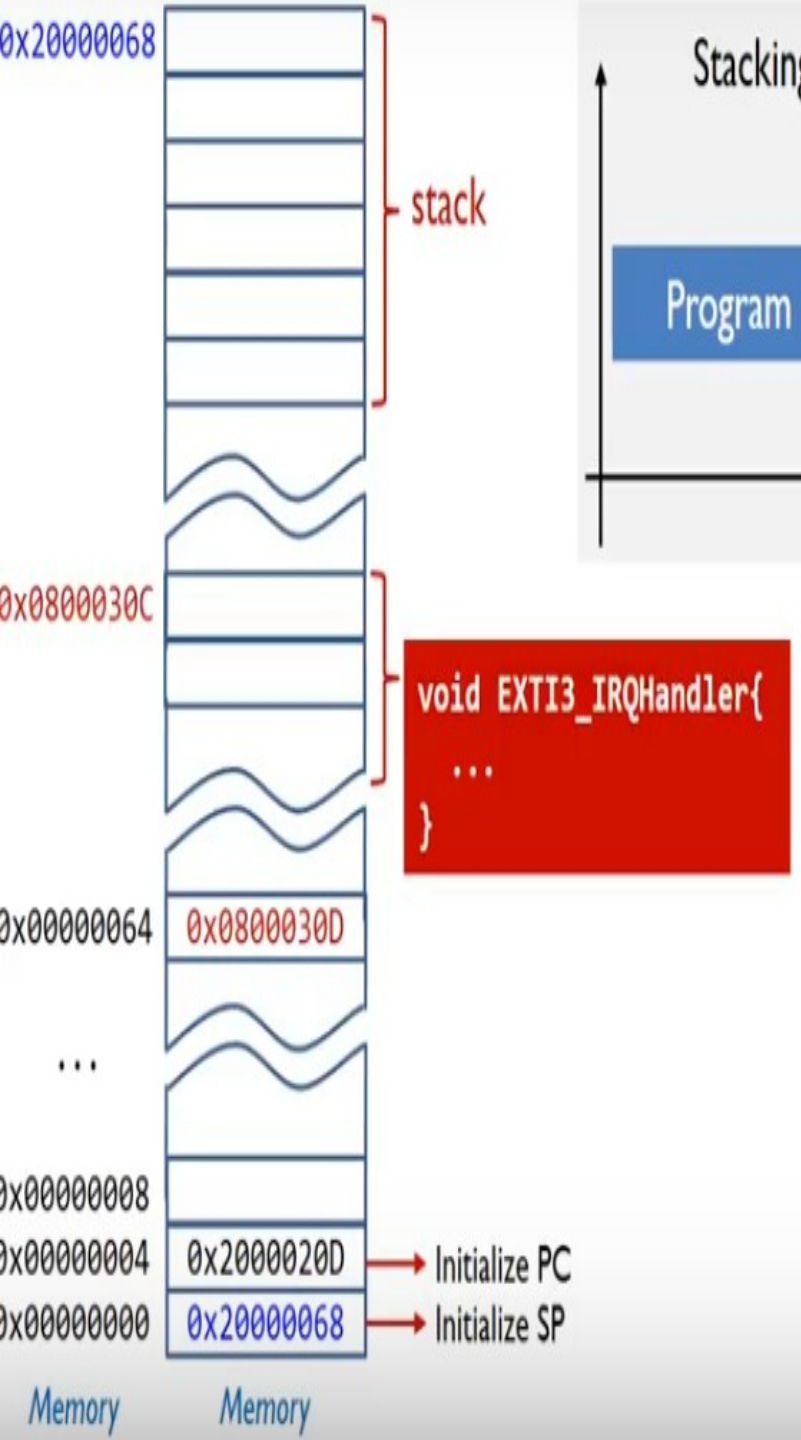
para los procesadores ARM Cortex, el bit 0 de la dirección de destino siempre debe ser 1.

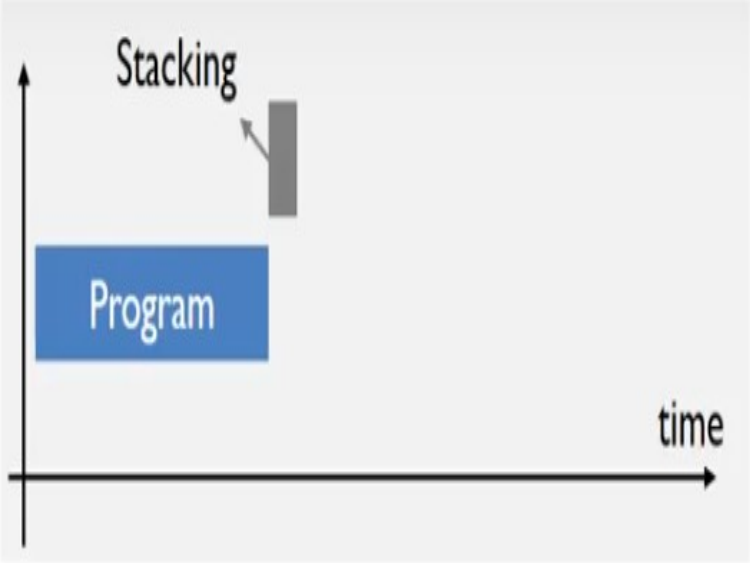
Single Interrupt

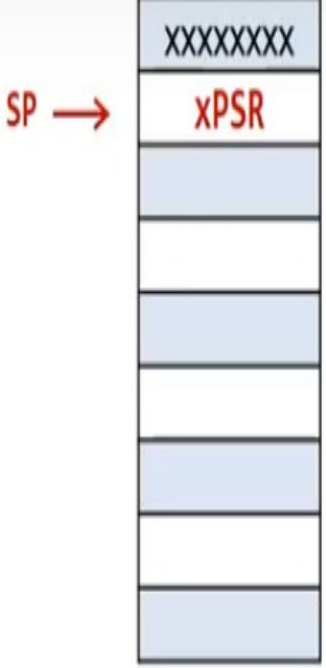




	EXTI3				
Interrupt Number	12	11	10	9	8
Enable Register	0	0	0	1	0
Active Register	0	0	0	0	0
Pending Register	0	0	0	0	0
Priority Register	3	4	7	2	8







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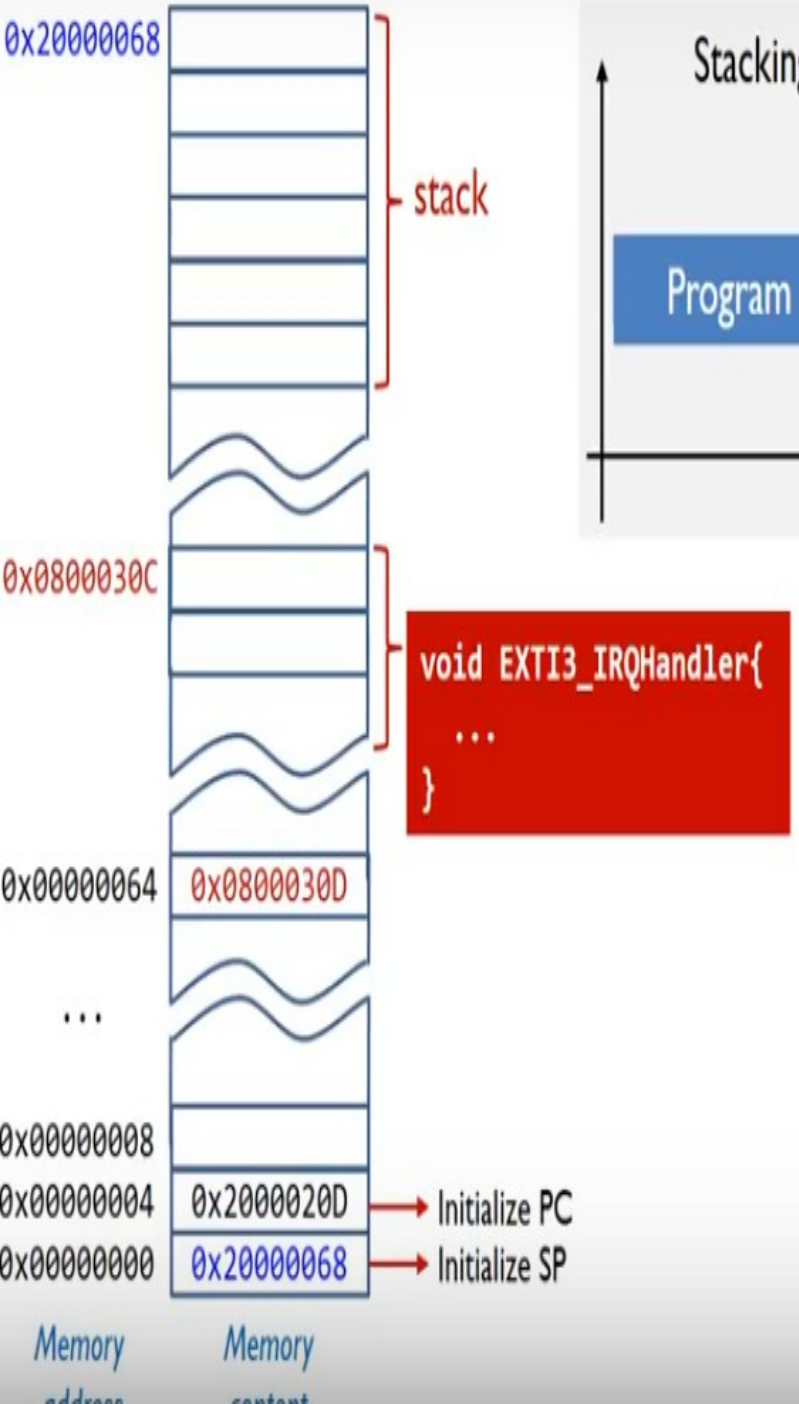

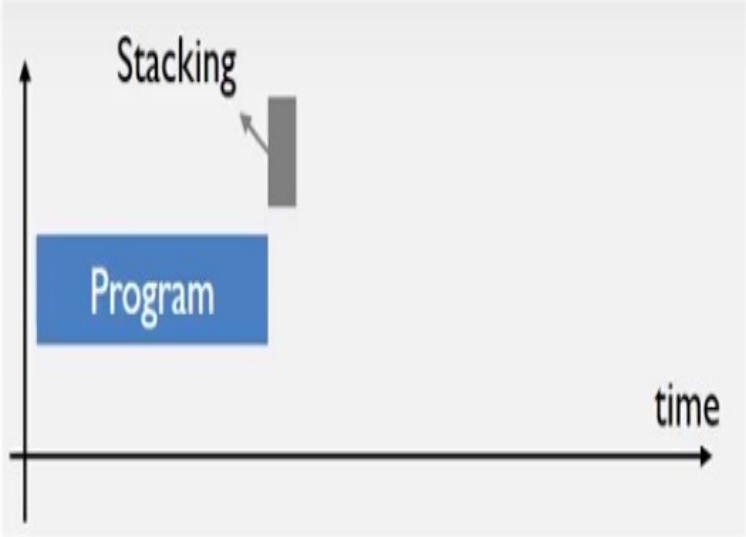

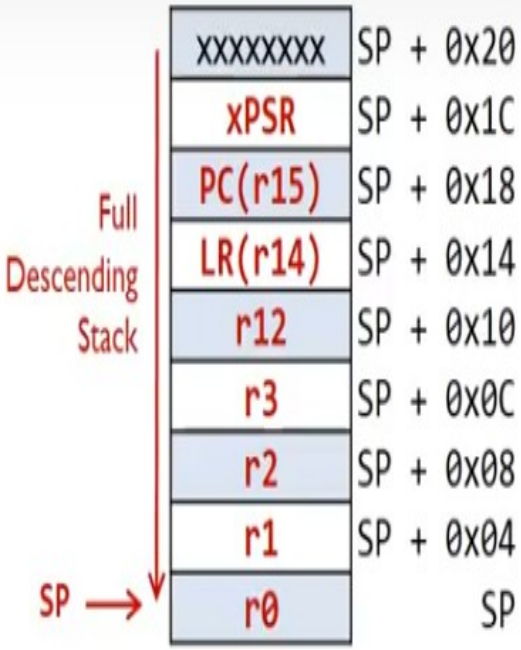

EXTI3

Interrupt Number	12	11	10	9	8
Enable Register	0	0	0	1	0
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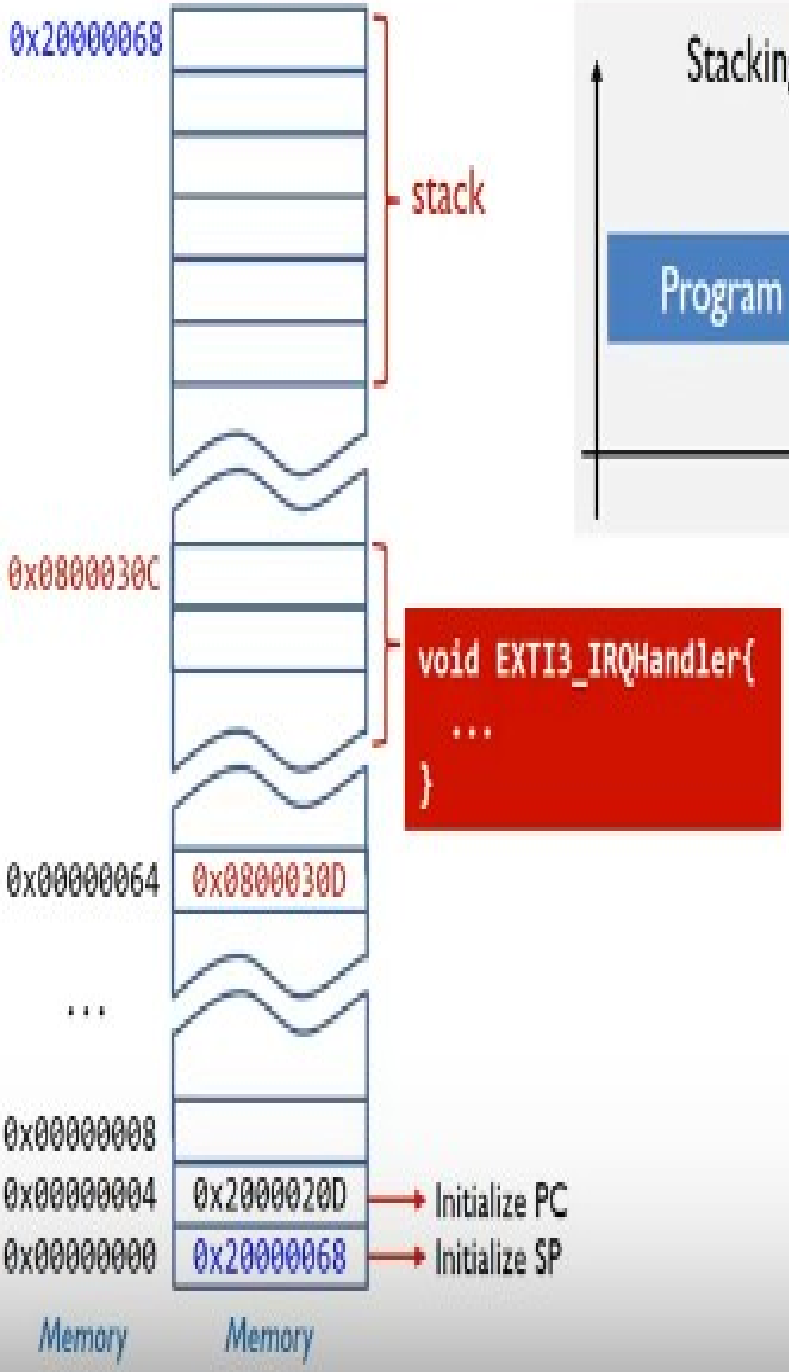

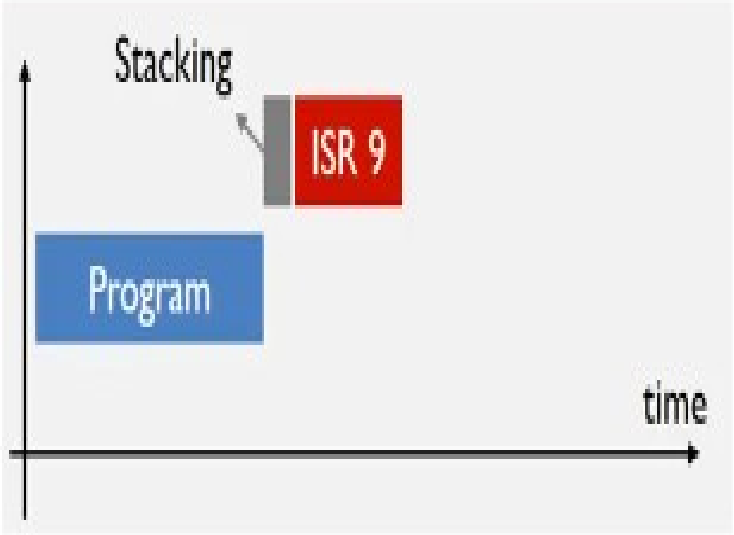

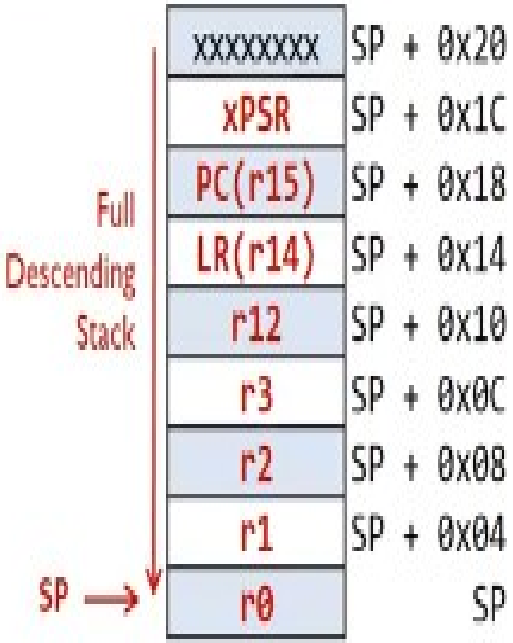

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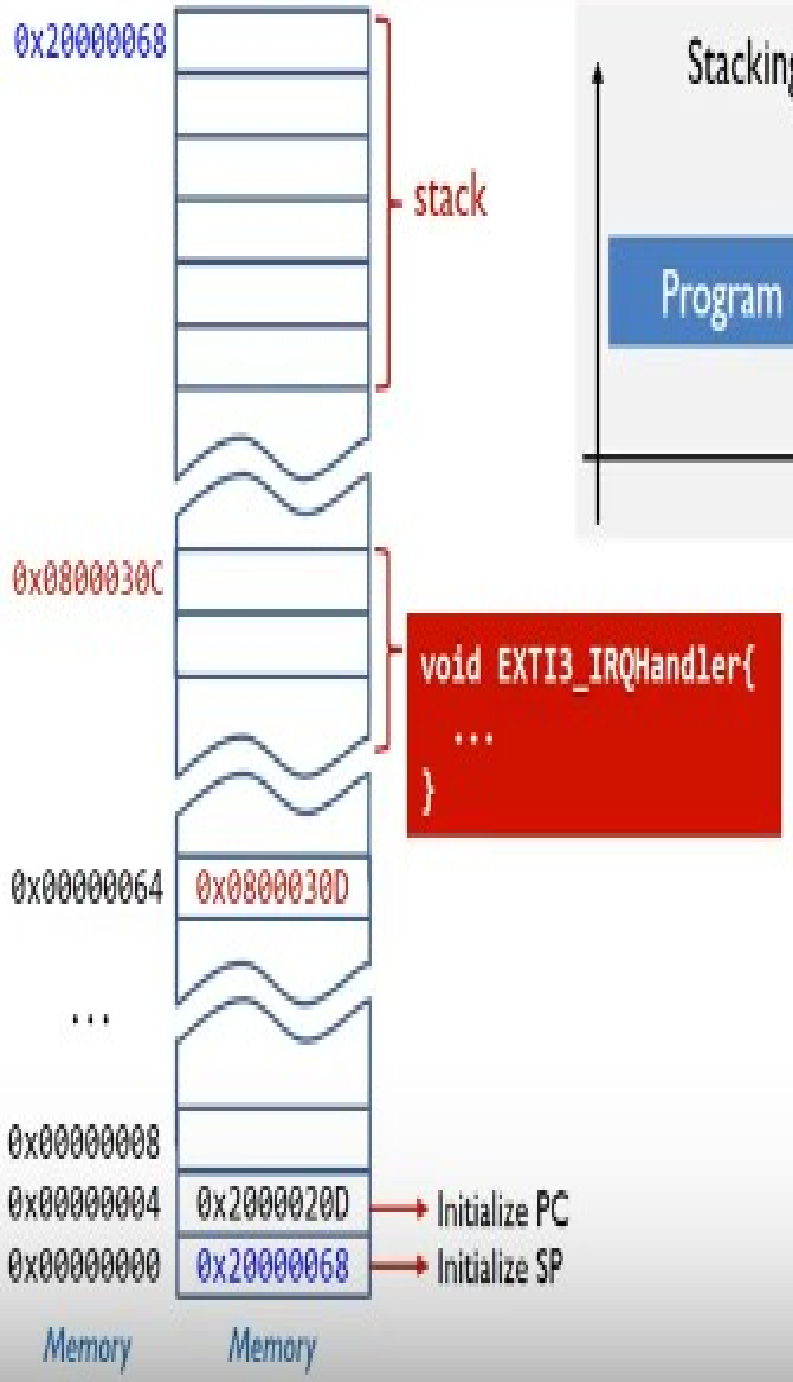

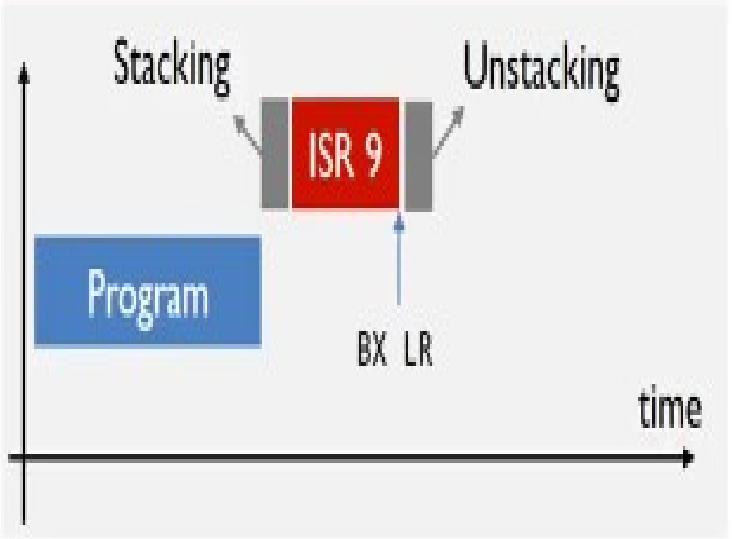

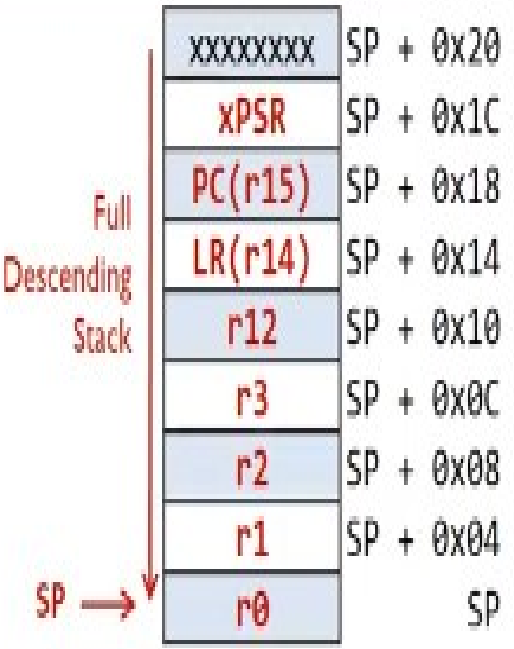

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Interrupt Number	12	11	10	9	8
Enable Register	0	0	0	1	0
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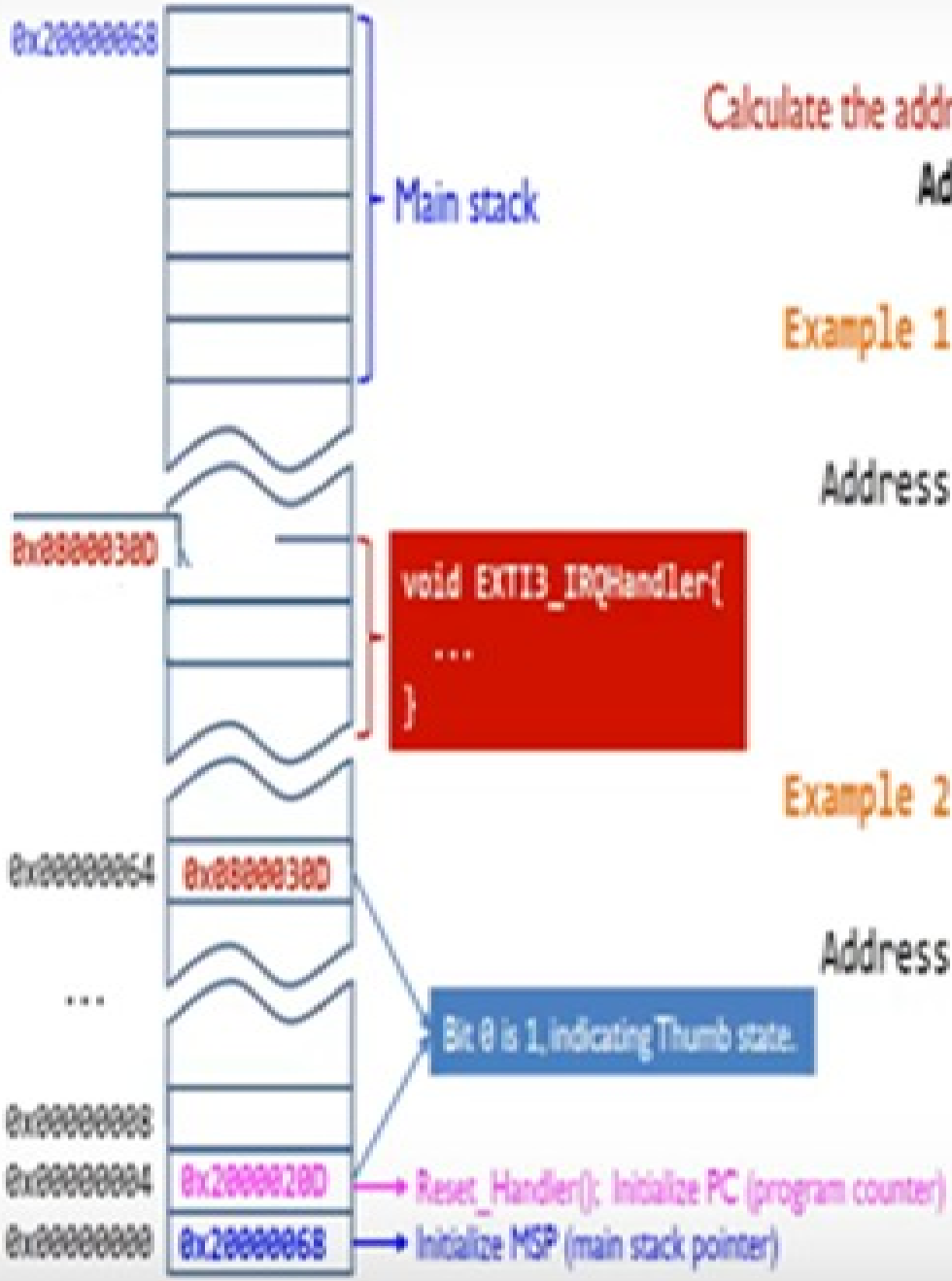

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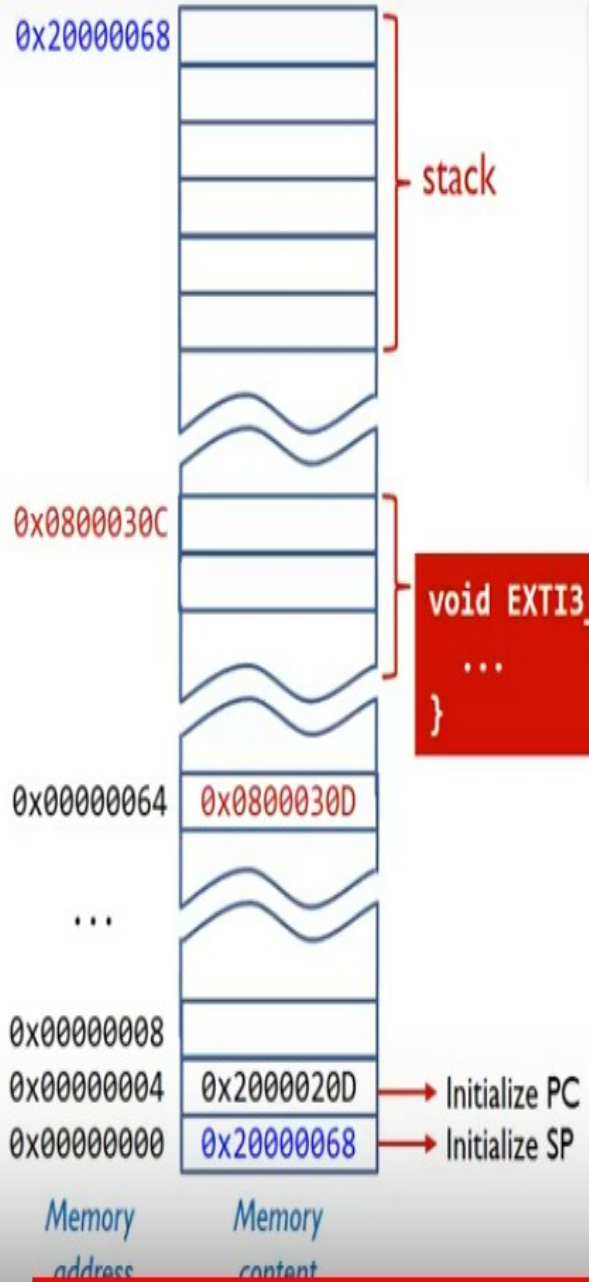
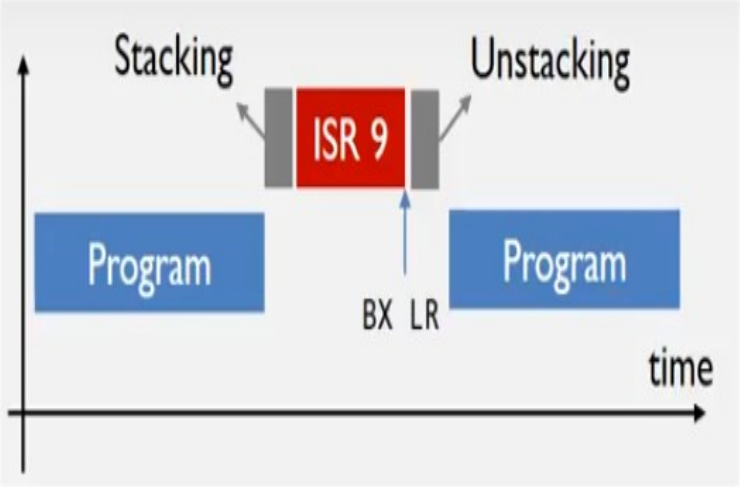
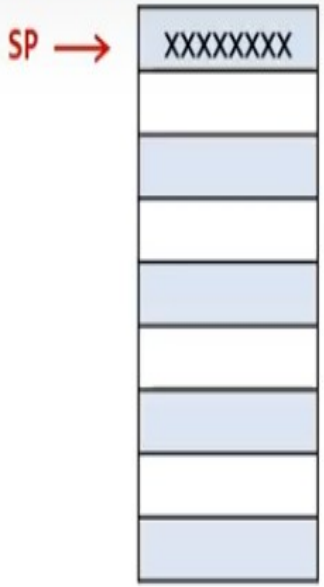
Single Interrupt

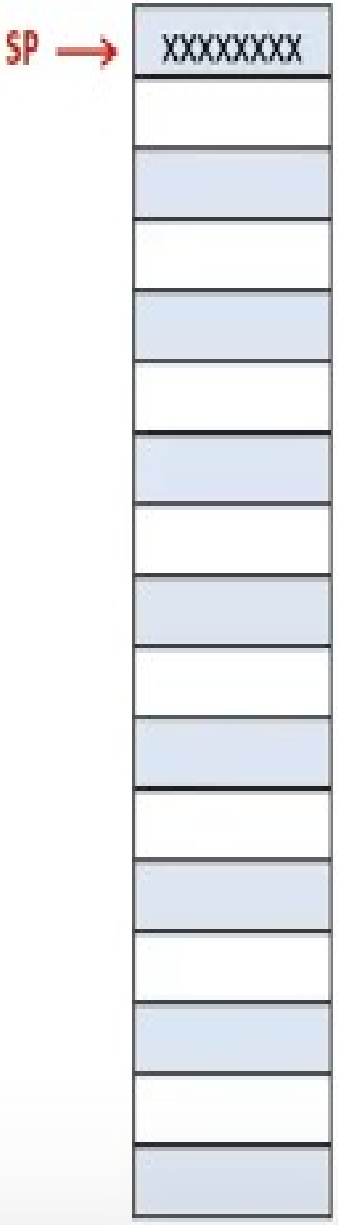




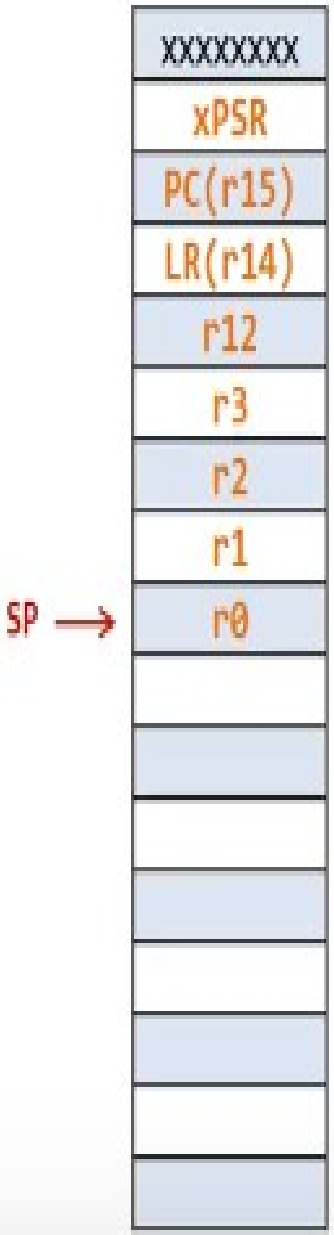


	EXTI3				
Interrupt Number	12	11	10	9	8
Enable Register	0	0	0	1	0
Active Register	0	0	0	0	0
Pending Register	0	0	0	0	0
Priority Register	3	4	7	2	8

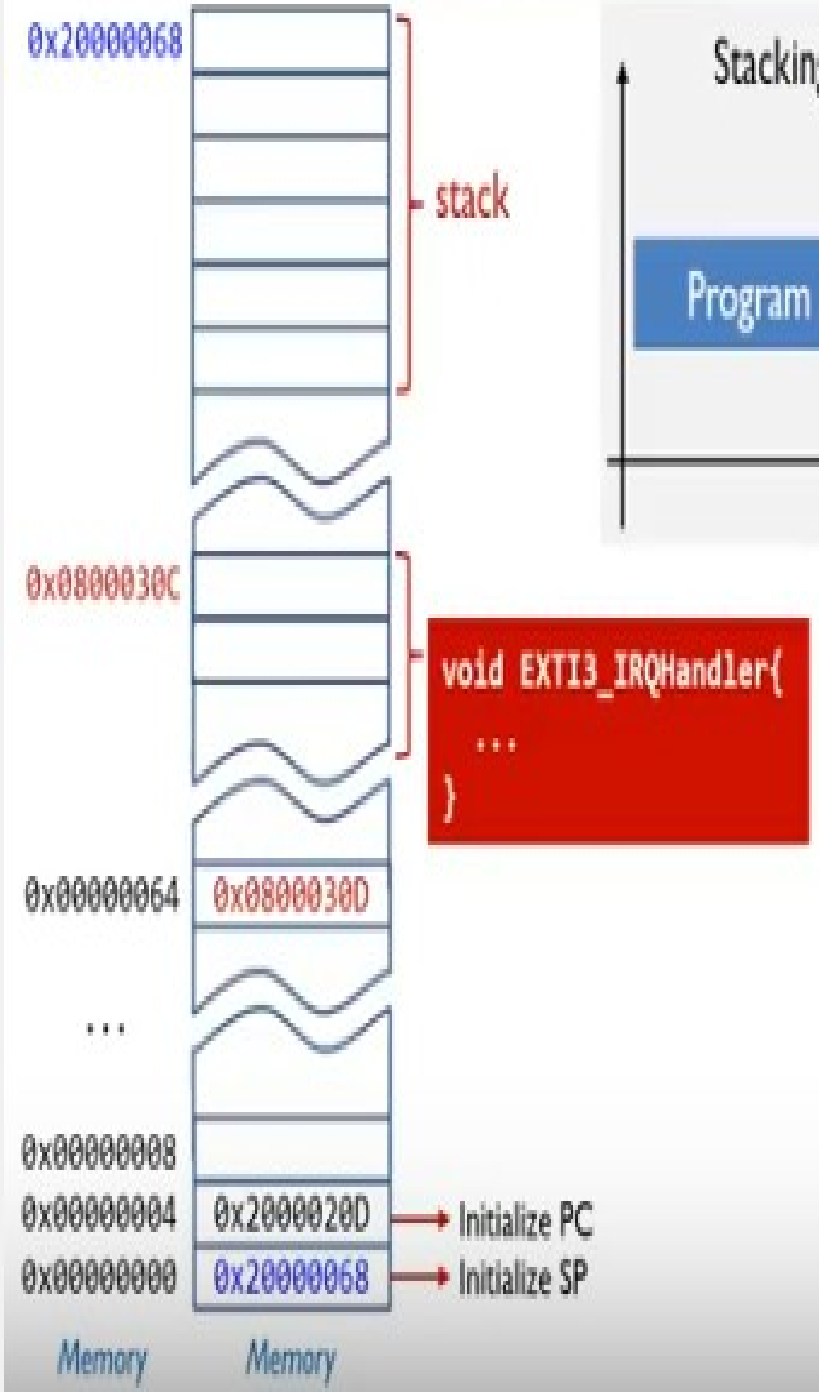
NESTED INTERRPTS

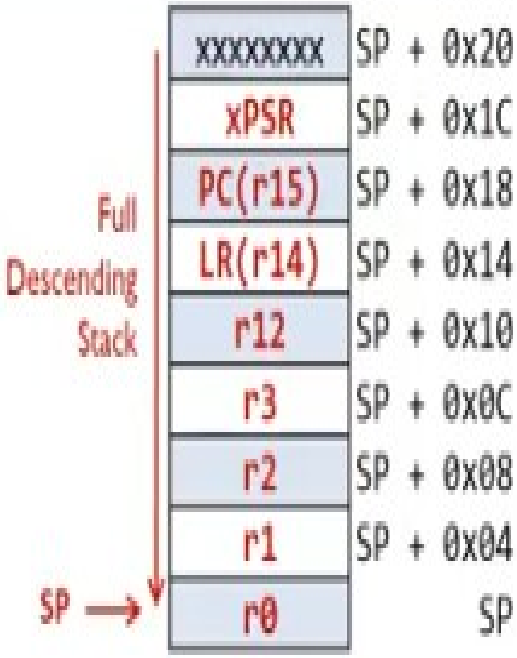


	DMA1_Channel2		EXTI3		
Interrupt Number	12	11	10	9	8
Enable Register	1	0	0	1	0
Active Register	0	0	0	0	0
Pending Register	0	0	0	0	0
Priority Register	3	4	7	5	3



	DMA1_Channel2			EXTI3	
Interrupt Number	12	11	10	9	8
Enable Register	1	0	0	1	0
Active Register	0	0	0	0	0
Pending Register	0	0	0	0	0
Priority Register	3	4	7	5	3





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Interrupt Number	12	11	10	9	8
Enable Register	0	0	0	1	0
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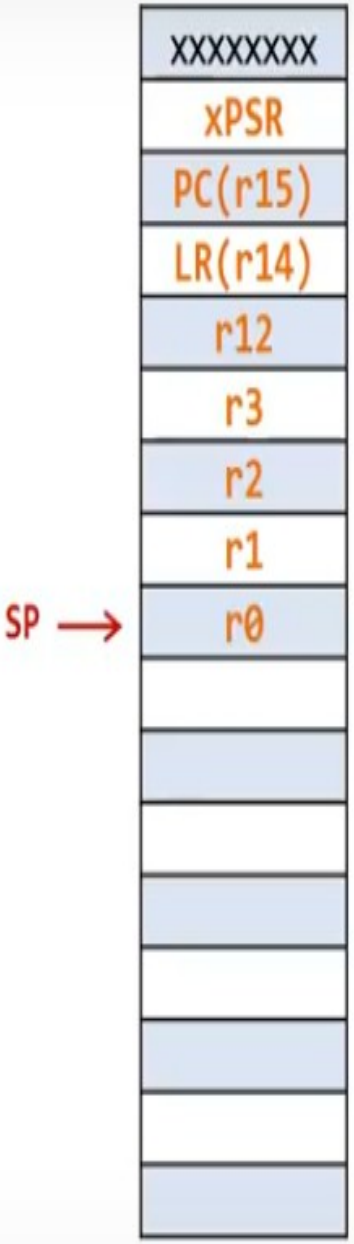
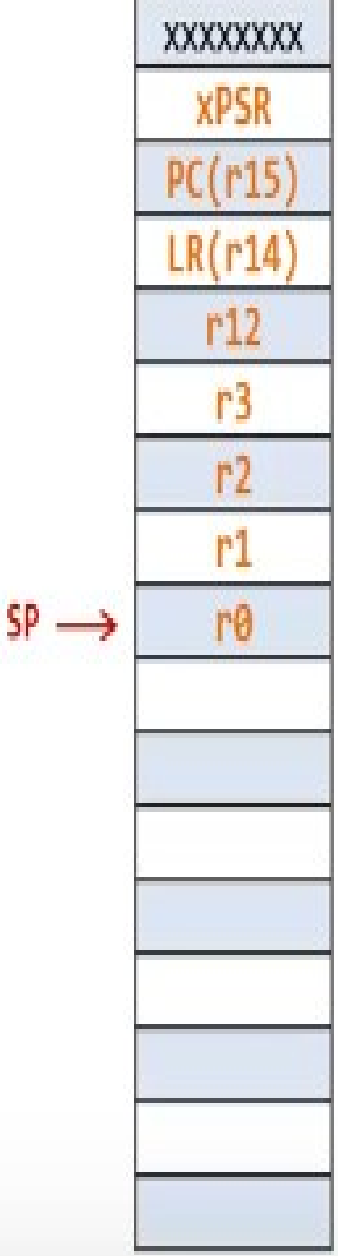
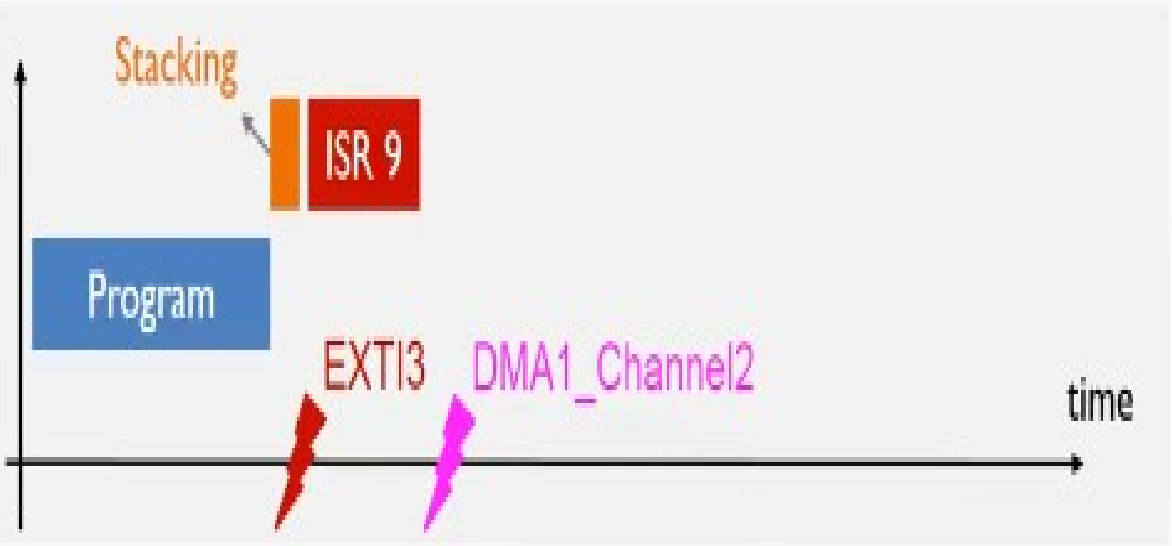
Pending Register	0	0	0	1	0
Priority Register	3	4	7	2	8

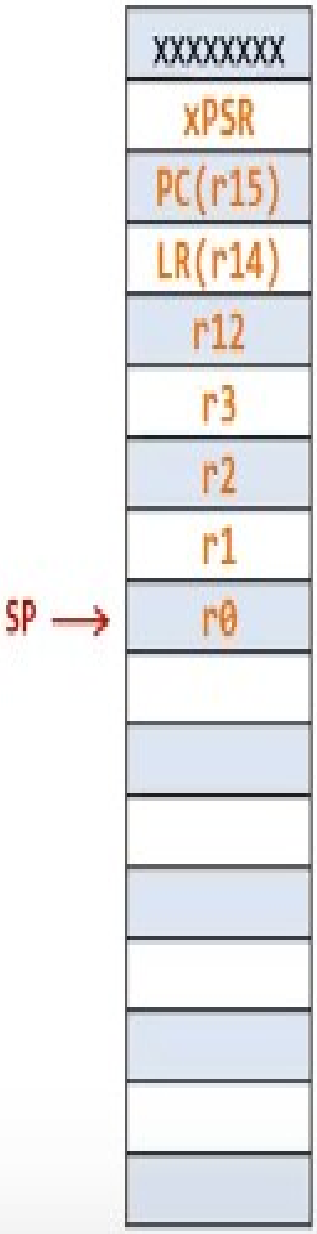


	DMA1_Channel2			EXTI3	
Interrupt Number	12	11	10	9	8
Enable Register	1	0	0	1	0
Active Register	0	0	0	1	0
Pending Register	0	0	0	0	0
Priority Register	3	4	7	5	3



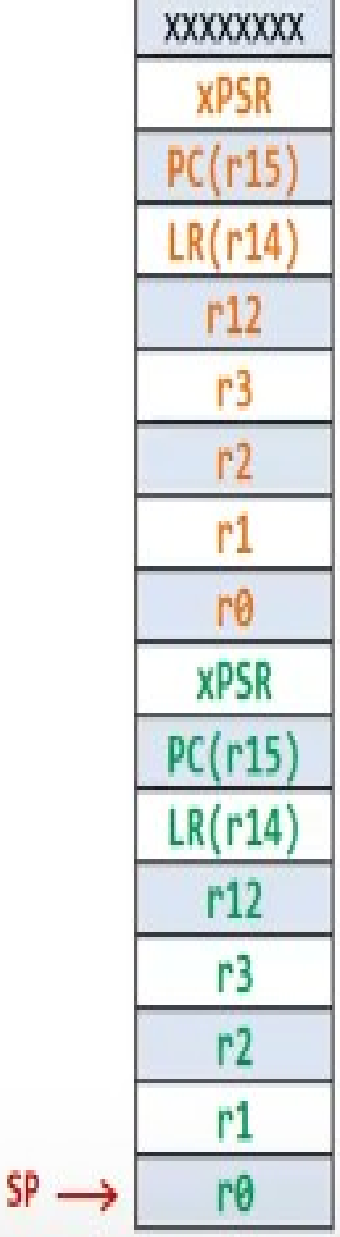
	DMA1_Channel2			EXTI3	
Interrupt Number	12	11	10	9	8
Enable Register	1	0	0	1	0
Active Register	0	0	0	1	0
Pending Register	0	0	0	0	0
Priority Register	3	4	7	5	3

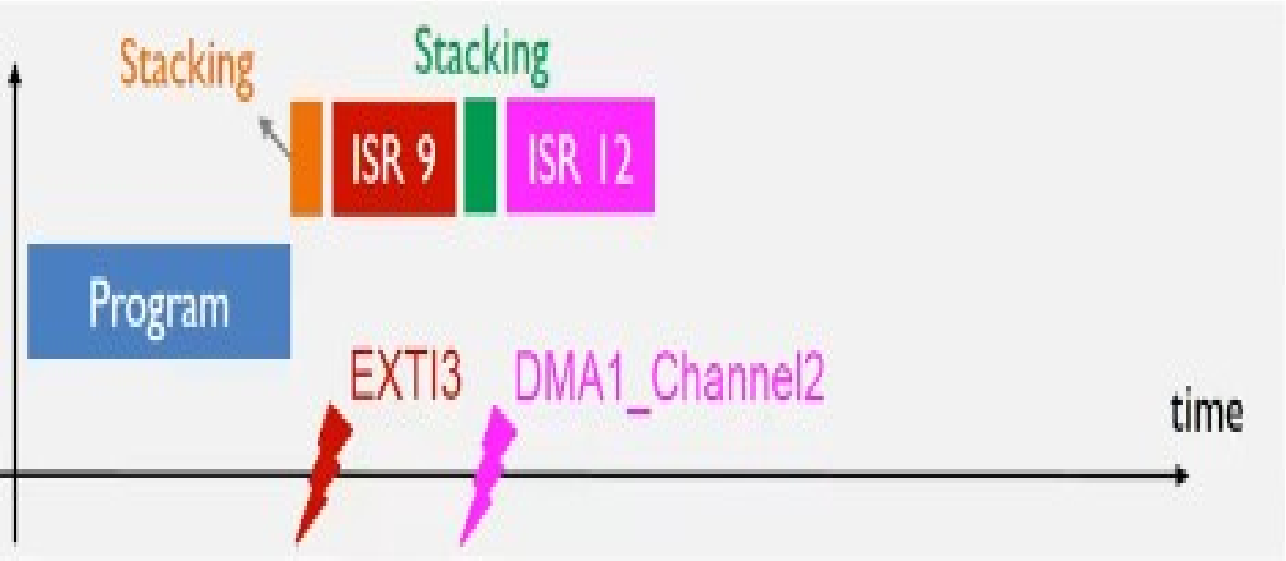




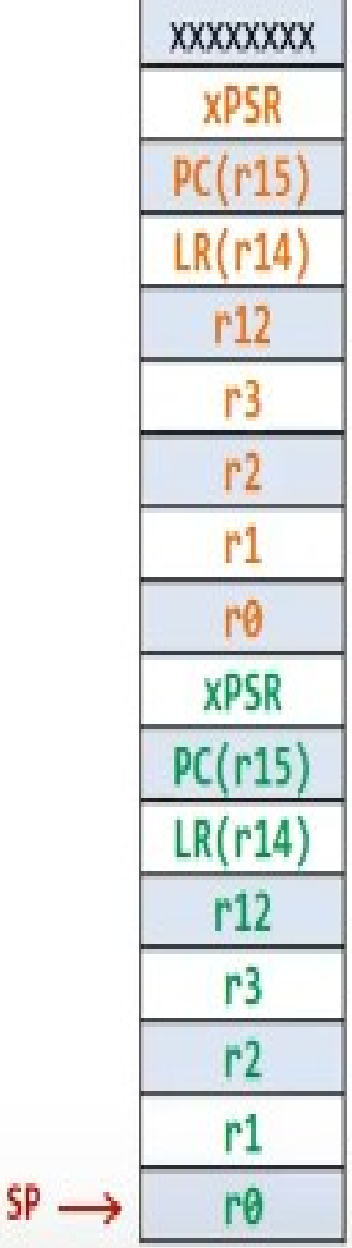
	DMA1_Channel2			EXTI3	
Interrupt Number	12	11	10	9	8
Enable Register	1	0	0	1	0
Active Register	0	0	0	1	0
Pending Register	1	0	0	0	0
Priority Register	3	4	7	5	3

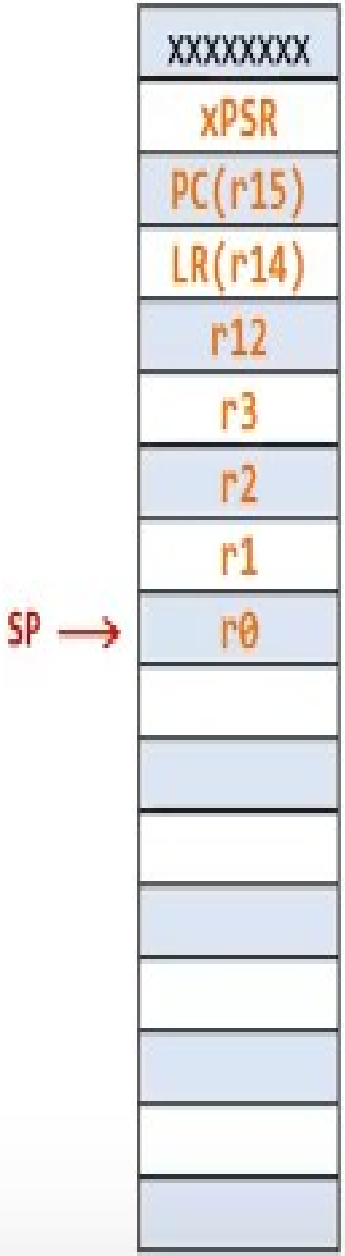
	DMA1_Channel2			EXTI3	
Interrupt Number	12	11	10	9	8
Enable Register	1	0	0	1	0
Active Register	1	0	0	1	0
Pending Register	0	0	0	0	0
Priority Register	3	4	7	5	3



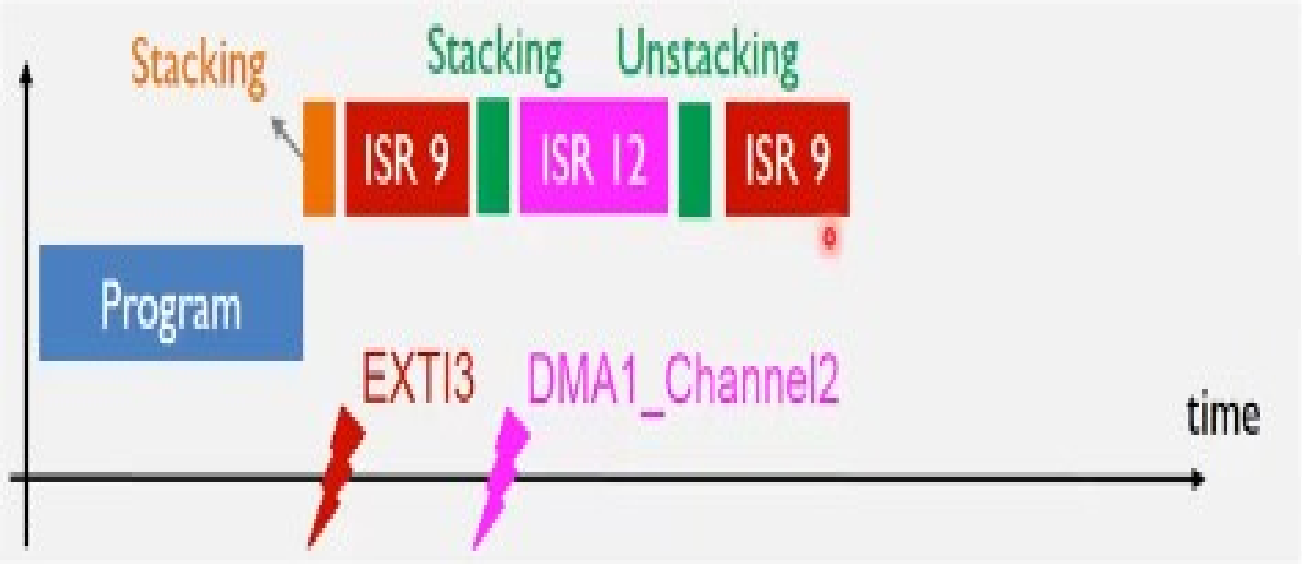


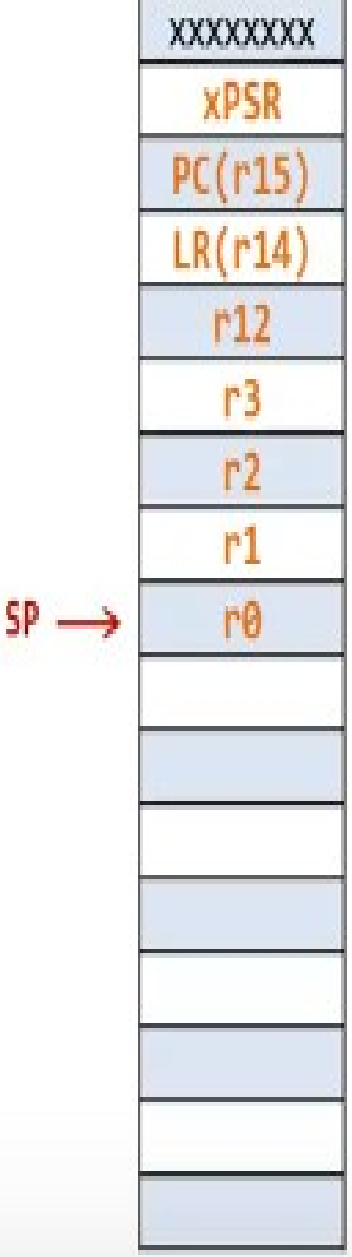
	DMA1_Channel2			EXTI3	
Interrupt Number	12	11	10	9	8
Enable Register	1	0	0	1	0
Active Register	1	0	0	1	0
Pending Register	0	0	0	0	0
Priority Register	3	4	7	5	3



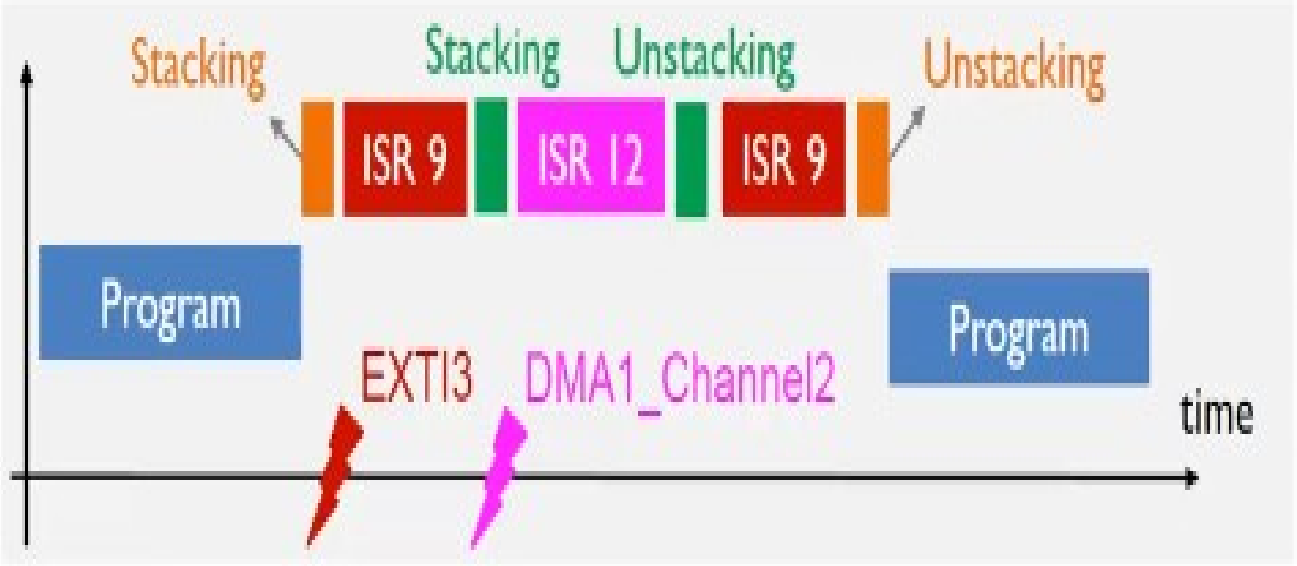


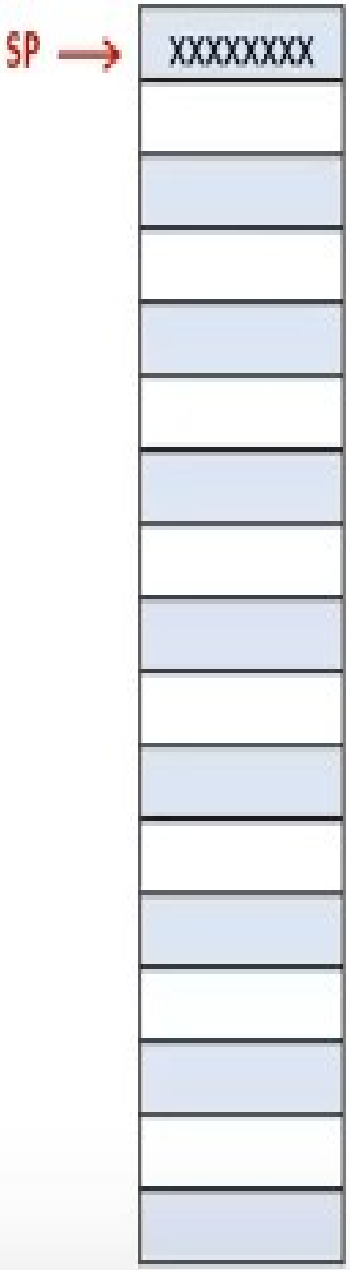
	DMA1_Channel2			EXTI3	
Interrupt Number	12	11	10	9	8
Enable Register	1	0	0	1	0
Active Register	0	0	0	1	0
Pending Register	0	0	0	0	0
Priority Register	3	4	7	5	3





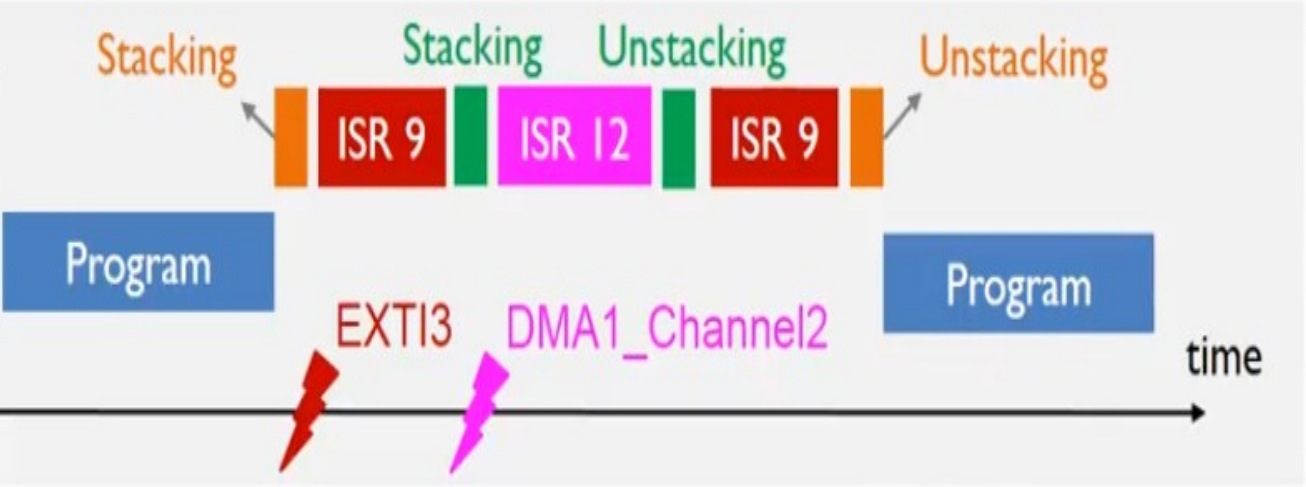
	DMA1_Channel2		EXTI3		
Interrupt Number	12	11	10	9	8
Enable Register	1	0	0	1	0
Active Register	0	0	0	1	0
Pending Register	0	0	0	0	0
Priority Register	3	4	7	5	3

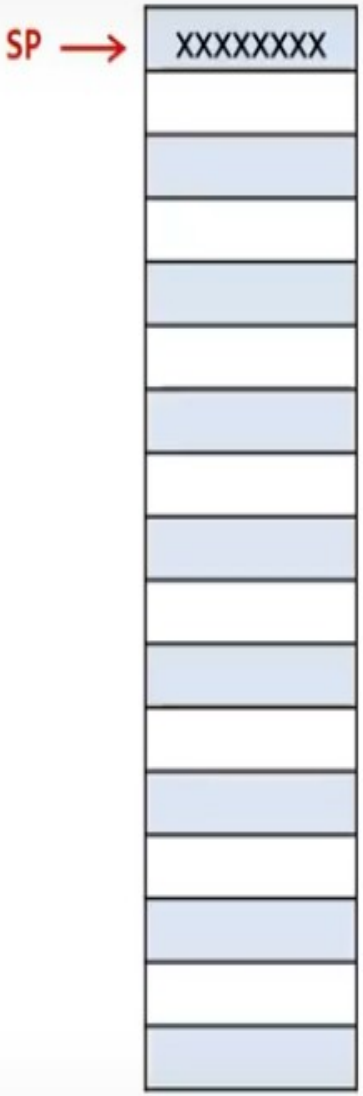




	DMA1_Channel2			EXTI3	
Interrupt Number	12	11	10	9	8
Enable Register	1	0	0	1	0
Active Register	0	0	0	0	0
Pending Register	0	0	0	0	0
Priority Register	3	4	7	5	3

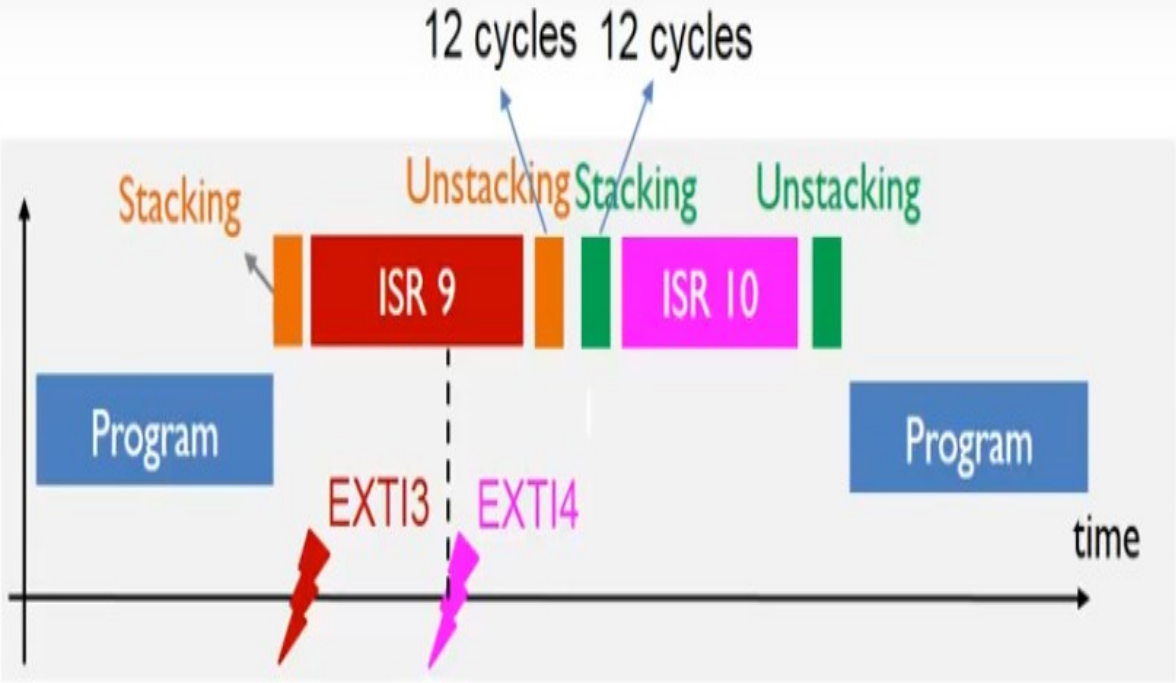
xample of Preemption

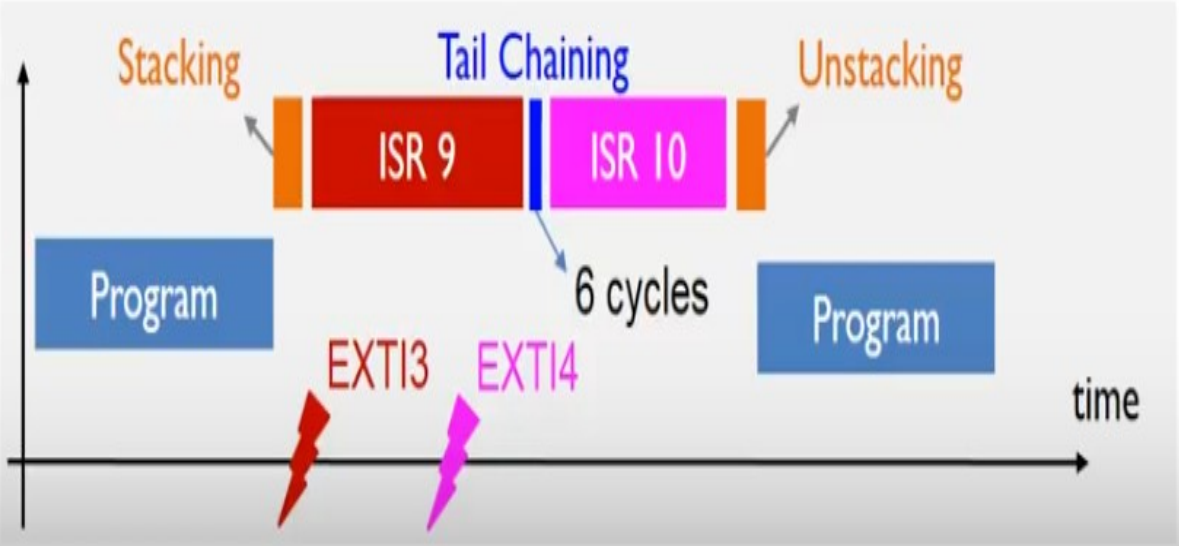


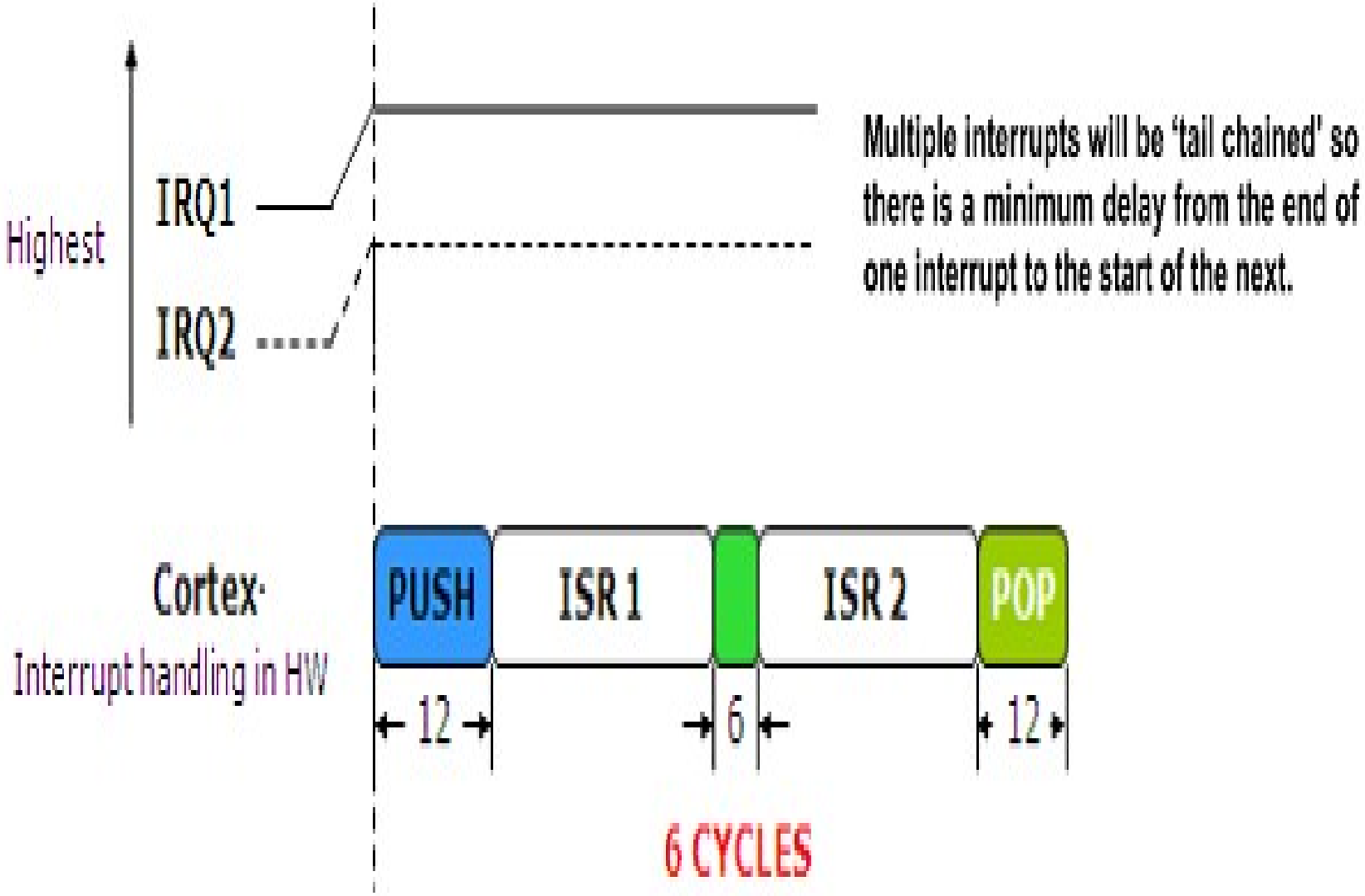


	DMA1_Channel2			EXTI3	
Interrupt Number	12	11	10	9	8
Enable Register	1	0	0	1	0
Active Register	0	0	0	0	0
Pending Register	0	0	0	0	0
Priority Register	3	4	7	5	3

- EXTI3 → ISR 9
- EXTI4 → ISR 10
- Suppose EXTI4 has less urgency than EXTI3.
  - EXTI4 has a higher numeric priority value than EXTI3.

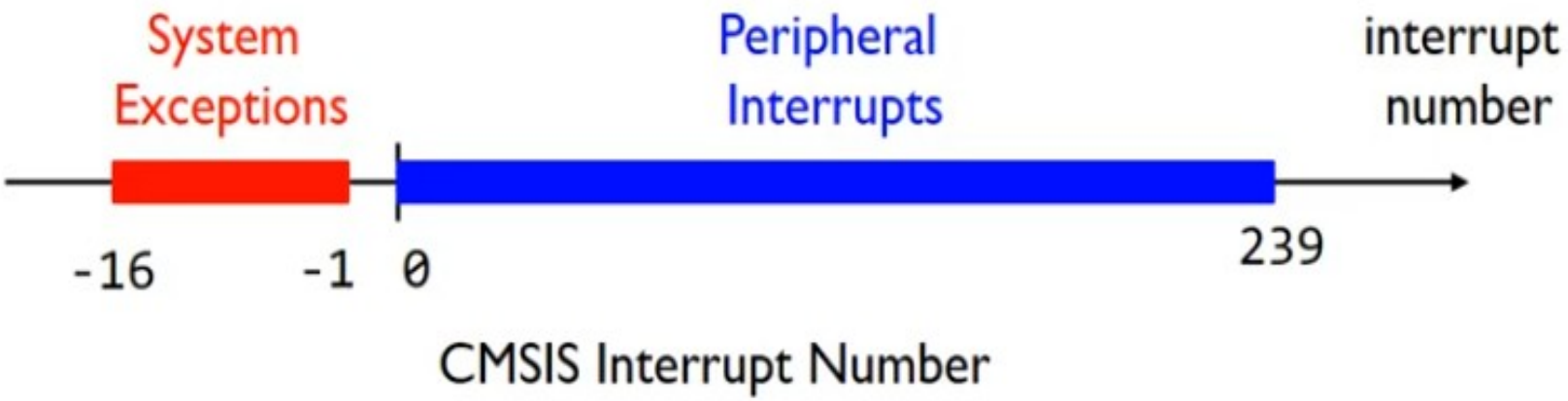





Highest
IRQ1
IRQ2
Multiple interrupts will be 'tail chained' so there is a minimum delay from the end of one interrupt to the start of the next.
Cortex-
Interrupt handling in HW
PUSH
ISR 1
ISR 2
POP
12
6
12
6 CYCLES

Interrupt Number

- Cortex-M supports up to 256 interrupts.



- First 16 are system exceptions
  - CMSIS defines their interrupt numbers as negative
  - Defined by ARM core
- The rest 240 are peripheral interrupts
  - Peripheral interrupt number starts with 0.
  - Defined by chip manufacturers.

Las primeras 16 interrupciones son interrupciones del sistema, también llamadas excepciones del sistema.

CMSIS define todas las excepciones del sistema mediante el uso de valores negativos.

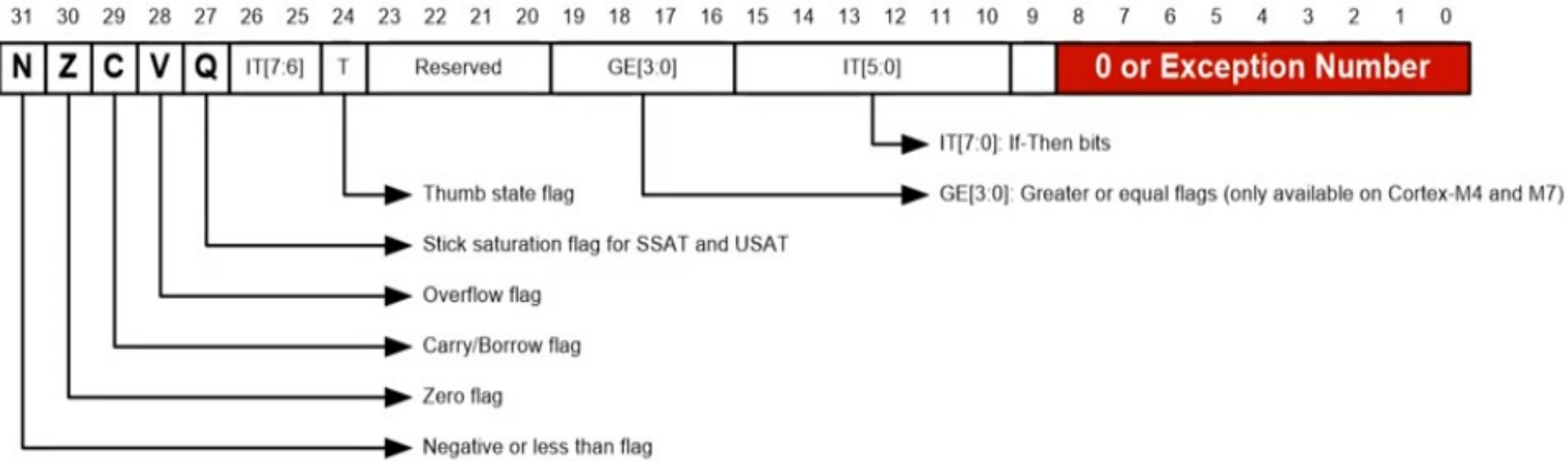
Las 240 interrupciones restantes son interrupciones periféricas, también llamadas excepciones que no son del sistema.

Interrupt Number in CMSIS *vs* in PSR

Interrupt number for CMSIS functions

```
NVIC_DisableIRQ (IRQn);              // Disable interrupt
NVIC_EnableIRQ (IRQn);               // Enable interrupt
NVIC_ClearingPending (IRQn);         // clear pending status
NVIC_SetPriority (IRQn, priority); // set priority level
```

Interrupt number in Program Status Register (PSR)



Interrupt Number in PSR = 16 + Interrupt Number for CMSIS

CMSIS Interrupt Number

```
******  Cortex-M4 System Exceptions ********************************************/
NonMaskableInt_IRQn    = -14,    /* 2 Cortex-M4 Non Maskable Interrupt                */
HardFault_IRQn         = -13,    /* 3 Cortex-M4 Hard Fault Interrupt                  */
MemoryManagement_IRQn  = -12,    /* 4 Cortex-M4 Memory Management Interrupt           */
BusFault_IRQn          = -11,    /* 5 Cortex-M4 Bus Fault Interrupt                   */
UsageFault_IRQn        = -10,    /* 6 Cortex-M4 Usage Fault Interrupt                 */
SVCall_IRQn            = -5,     /* 11 Cortex-M4 SV Call Interrupt                    */
DebugMonitor_IRQn      = -4,     /* 12 Cortex-M4 Debug Monitor Interrupt              */
PendSV_IRQn            = -2,     /* 14 Cortex-M4 Pend SV Interrupt                    */
SysTick_IRQn           = -1,     /* 15 Cortex-M4 System Tick Interrupt                */
******  Peripheral Interrupt Numbers *******************************************/
WWDG_IRQn              = 0,      /* Window WatchDog Interrupt                         */
PVD_PVM_IRQn           = 1,      /* PVD/PVM1,2,3,4 through EXTI Line detection Interrupts */
TAMP_STAMP_IRQn        = 2,      /* Tamper and TimeStamp interrupts through the EXTI line */
RTC_WKUP_IRQn          = 3,      /* RTC Wakeup interrupt through the EXTI line        */
FLASH_IRQn             = 4,      /* FLASH global Interrupt                            */
RCC_IRQn               = 5,      /* RCC global Interrupt                              */
EXTI0_IRQn             = 6,      /* EXTI Line0 Interrupt                              */
...
```

System Exceptions

Peripheral Interrupts

stm32l476xx.h

Enable an Interrupt

- Enable a system exception
  - Some are always enabled (cannot be disabled)
  - No centralized registers for enabling/disabling
  - Each are control by its corresponding components, such as SysTick module

- Enable a peripheral interrupt
  - Centralized register arrays for enabling/disabling
  - **ISER** registers for enabling
  - **ICER** registers for disabling

Específicamente, los registros ISER se utilizan para habilitar las interrupciones periféricas.

Los registros ICER se utilizan para deshabilitar las interrupciones periféricas.

Enabling Peripheral Interrupts

Interrupt Set Enable Register 0 (ISER0)

Interrupt Number	Enable Bit	Peripheral
31	0	I2C1_EV
30	0	TIM4
29	0	TIM3
28	0	TIM2
27	0	TIM11
26	0	TIM10
25	0	TIM9
24	0	LCD
23	0	EXTI9_5
22	0	COMP
21	0	DAC
20	0	USB_LP
19	0	USB_HP
18	0	ADC1
17	0	DMA1_CH7
16	0	DMA1_CH6
15	0	DMA1_CH5
14	0	DMA1_CH4
13	0	DMA1_CH3
12	0	DMA1_CH2
11	0	DMA1_CH1
10	0	EXTI4
9	0	EXTI3
8	0	EXTI2
7	0	EXTI1
6	0	EXTI0
5	0	RCC
4	0	FLASH
3	0	RTC_WKUP
2	0	TAMPER_STAMP
1	0	PVD
0	0	WWDG

Interrupt Set Enable Register 1 (ISER1)

Address of ISER1 = Address of ISER0 + 4

Interrupt Number	Enable Bit	Peripheral
31	0	
30	0	
29	0	
28	0	
27	0	
26	0	
25	0	
24	0	
23	0	
22	0	
21	0	
20	0	
19	0	
18	0	
17	0	
16	0	
15	0	
14	0	
13	0	
12	1	
11	0	
10	0	
9	0	
8	0	
7	0	
6	0	
5	0	
4	0	
3	0	
2	0	
1	0	
0	0	
44		TIM7
43		TIM6
42		USB_FS_WKUP
41		RTC_Alarm
40		EXTI15_10
39		USART3
38		USART2
37		USART1
36		SPI2
35		SPI1
34		I2C2_ER
33		I2C2_EV
32		I2C1_ER

```
TIM7_IRQn = 44

NVIC->ISER[1] = 1 << 12;      // Enable Timer 7 interrupt
```

Disabling Peripheral Interrupts

Interrupt Clear Enable Register 0 (ICER0)

	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
lear Enable Bit	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
terrupt Number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	I2C1_EV	TIM4	TIM3	TIM2	TIM11	TIM10	TIM9	LCD	EXTI9_5	COMP	DAC	USB_LP	USB_HP	ADC1	DMA1_CH7	DMA1_CH6	DMA1_CH5	DMA1_CH4	DMA1_CH3	DMA1_CH2	DMA1_CH1	EXTI4	EXTI3	EXTI2	EXTI1	EXTI0	RCC	FLASH	RTC_WKUP	TAMPER_STAMP	PVD	WWDG

Interrupt Clear Enable Register 1 (ICER1)

Address of ICER1 = Address of ISER0 + 4

	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
lear Enable Bit	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
terrupt Number																				44	43	42	41	40	39	38	37	36	35	34	33	32
																				TIM7	TIM6	USB_FS_WKUP	RTC_Alarm	EXTI15_10	USART3	USART2	USART1	SPI2	SPI1	I2C2_ER	I2C2_EV	I2C1_ER

```
TIM7_IRQn = 44

NVIC->ICER[1] = 1 << 12;      // Diable Timer 7 interrupt
```

Disable/Enable Peripheral Interrupts

- For all peripheral interrupts: `IRQn ≥ 0`
- Method 1:
  - `NVIC_EnableIRQ (IRQn);  // Enable interrupt`
  - `NVIC_DisableIRQ (IRQn); // Disable interrupt`
- Method 2:
  - Enable:
    - `NVIC->ISER[ IRQn / 32] = 1 << (IRQn % 32);`
    - Better solution:
    - `NVIC->ISER[ IRQn >> 5] = 1 << (IRQn & 0x1F);`
  - Disable:
    - `NVIC->ICER[ IRQn >> 5] = 1 << (IRQn & 0x1F);`

Interrupt Priority

- Inverse Relationship:
  - Lower priority value means higher urgency.
    - Priority of Interrupt A = 5,
    - Priority of Interrupt B = 2,
    - B has a higher priority/urgency than A.

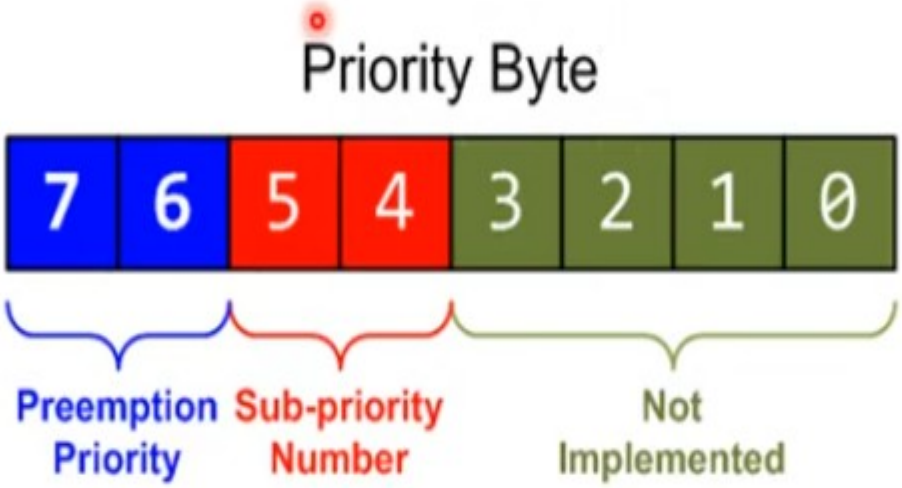
- Fixed priority for Reset, HardFault, and NMI.

Exception	IRQn	Priority
Reset	N/A	-3 (the highest)
Non-maskable Interrupt (NMI)	-14	-2 (2nd highest)
Hard Fault	-13	-1

- Adjustable for all the other interrupts

Interrupt Priority

- Interrupt priority is configured by Interrupt Priority Register (IP)
- Each priority consists of two fields, including preempt priority number and sub-priority number.
  - The preempt priority number defines the priority for preemption.
  - The sub-priority number determines the order when multiple interrupts are pending with the same preempt priority number.



default setting

Table 51. NVIC register map

Name	Description	Access	Reset value	Address
ISER0 to ISER1	Interrupt Set-Enable Registers. These 2 registers allow enabling interrupts and reading back the interrupt enables for specific peripheral functions.	RW	0	ISER0 - 0xE000 E100 ISER1 - 0xE000 E104
ICER0 to ICER1	Interrupt Clear-Enable Registers. These 2 registers allow disabling interrupts and reading back the interrupt enables for specific peripheral functions.	RW	0	ICER0 - 0xE000 E180 ICER1 - 0xE000 E184
ISPR0 to ISPR1	Interrupt Set-Pending Registers. These 2 registers allow changing the interrupt state to pending and reading back the interrupt pending state for specific peripheral functions.	RW	0	ISPR0 - 0xE000 E200 ISPR1 - 0xE000 E204
ICPR0 to ICPR1	Interrupt Clear-Pending Registers. These 2 registers allow changing the interrupt state to not pending and reading back the interrupt pending state for specific peripheral functions.	RW	0	ICPR0 - 0xE000 E280 ICPR1 - 0xE000 E284
IABR0 to IABR1	Interrupt Active Bit Registers. These 2 registers allow reading the current interrupt active state for specific peripheral functions.	RO	0	IABR0 - 0xE000 E300 IABR1 - 0xE000 E304
IPR0 to IPR8	Interrupt Priority Registers. These 9 registers allow assigning a priority to each interrupt. Each register contains the 5-bit priority fields for 4 interrupts.	RW	0	IPR0 - 0xE000 E400 IPR1 - 0xE000 E404 IPR2 - 0xE000 E408 IPR3 - 0xE000 E40C IPR4 - 0xE000 E410 IPR5 - 0xE000 E414 IPR6 - 0xE000 E418 IPR7 - 0xE000 E41C IPR8 - 0xE000 E420

Table 67. Interrupt Priority Register 5 (IPR5 - 0xE000 E414)

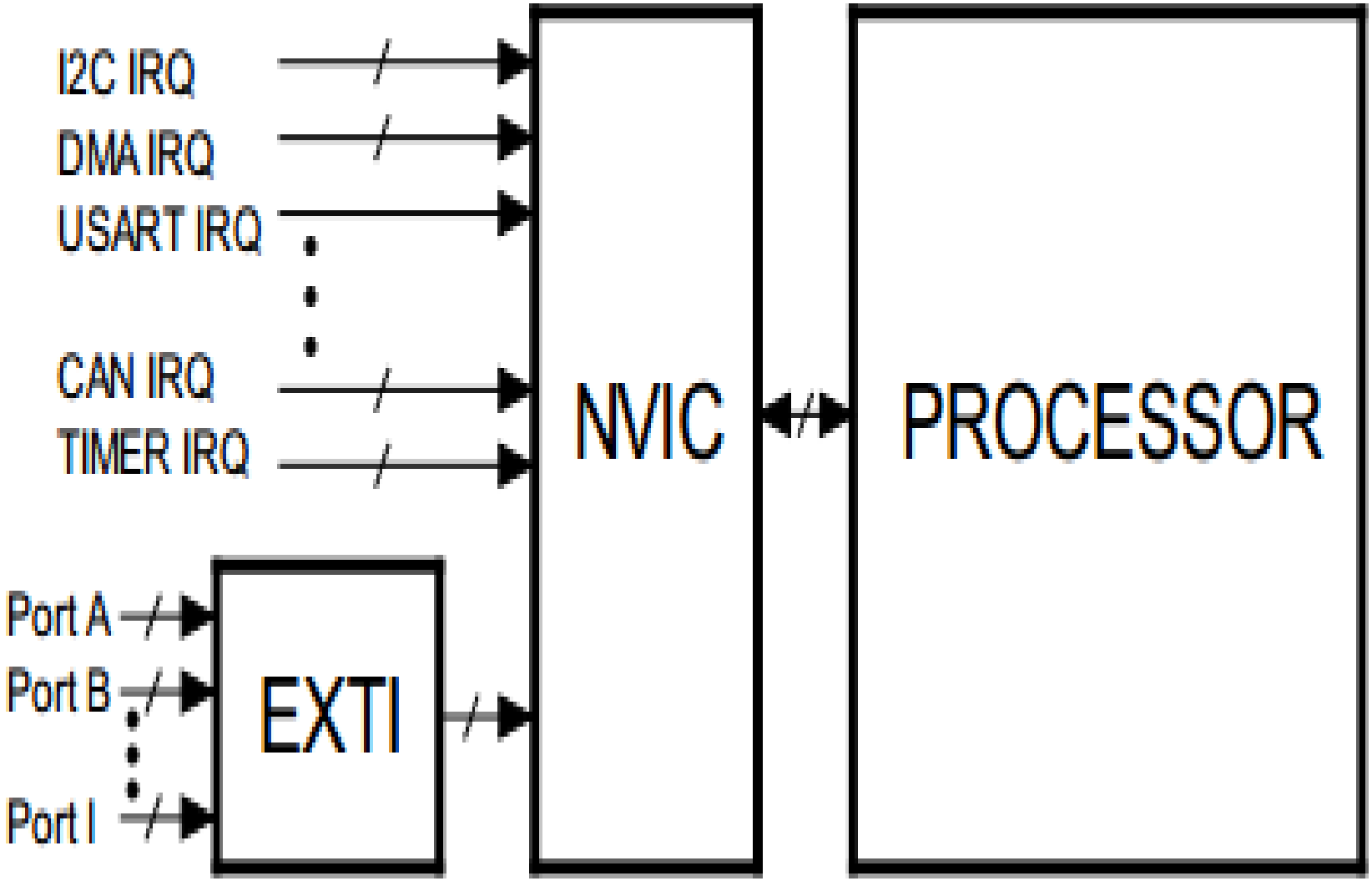
Bit	Name	Function
2:0	Unimplemented	These bits ignore writes, and read as 0.
7:3	IP_EINT2	External Interrupt 2 Interrupt Priority. 0 = highest priority. 31 (0x1F) = lowest priority.
10:8	Unimplemented	These bits ignore writes, and read as 0.
15:11	IP_EINT3	External Interrupt 3 Interrupt Priority. See functional description for bits 7-3.
18:16	Unimplemented	These bits ignore writes, and read as 0.
23:19	IP_ADC	ADC Interrupt Priority. See functional description for bits 7-3.
26:24	Unimplemented	These bits ignore writes, and read as 0.
31:27	IP_BOD	BOD Interrupt Priority. See functional description for bits 7-3.

				IPR7 - 0xE000 E41C IPR8 - 0xE000 E420
STIR	Software Trigger Interrupt Register. This register allows software to generate an interrupt.	WO	0	STIR - 0xE000 EF00

Table 71. Software Trigger Interrupt Register (STIR - 0xE000 EF00)

Bit	Name	Function
8:0	INTID	Writing a value to this field generates an interrupt for the specified the interrupt number (see Table 50). The range allowed for the LPC176x/5x is 0 to 111.
31:9	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.

INTERRUPCIONES EXTERNAS


I2C IRQ
DMA IRQ
USART IRQ
CAN IRQ
TIMER IRQ
NVIC
PROCESSOR
Port A
Port B
Port I
EXTI

El LPC17xx incluye cuatro entradas de interrupción externas como funciones de pin seleccionables. Además, las interrupciones externas tiene la capacidad de reactivar la CPU desde el modo de apagado y Despertar desde modos de energía reducida.

PINES

Table 6. Pin summary

Pin name	Pin direction	Pin description
EINT0	Input	**External Interrupt Input 0** - An active low/high level or falling/rising edge general purpose interrupt input. This pin may be used to wake up the processor from Sleep, Deep-sleep, or Power-down modes.
EINT1	Input	**External Interrupt Input 1** - See the EINT0 description above.
EINT2	Input	**External Interrupt Input 2** - See the EINT0 description above.
EINT3	Input	**External Interrupt Input 3** - See the EINT0 description above.
$\overline{\text{RESET}}$	Input	**External Reset input** - A LOW on this pin resets the chip, causing I/O ports and peripherals to take on their default states, and the processor to begin execution at address 0x0000 0000.

					I	**RI1** — Ring Indicator input for UART1.
					O	**TRACECLK** — Trace Clock.
P2[7]/RD2/ RTS1	66	E9	J4	[1]	I/O	**P2[7]** — General purpose digital input/output pin.
					I	**RD2** — CAN2 receiver input. (LPC1769/68/66/65/64 only).
					O	**RTS1** — Request to Send output for UART1. Can also be configured to be an RS-485/EIA-485 output enable signal.
P2[8]/TD2/ TXD2	65	E10	H5	[1]	I/O	**P2[8]** — General purpose digital input/output pin.
					O	**TD2** — CAN2 transmitter output. (LPC1769/68/66/65/64 only).
					O	**TXD2** — Transmitter output for UART2.
P2[9]/ USB_CONNECT/ RXD2	64	F7	K5	[1]	I/O	**P2[9]** — General purpose digital input/output pin.
					O	**USB_CONNECT** — Signal used to switch an external 1.5 kΩ resistor under software control. Used with the SoftConnect USB feature. (LPC1769/68/66/65/64 only).
					I	**RXD2** — Receiver input for UART2.
P2[10]/$\overline{\text{EINT0}}$/NMI	53	J10	K9	[6]	I/O	**P2[10]** — General purpose digital input/output pin. A LOW level on this pin during reset starts the ISP command handler.
					I	**$\overline{\text{EINT0}}$** — External interrupt 0 input.
					I	**NMI** — Non-maskable interrupt input.

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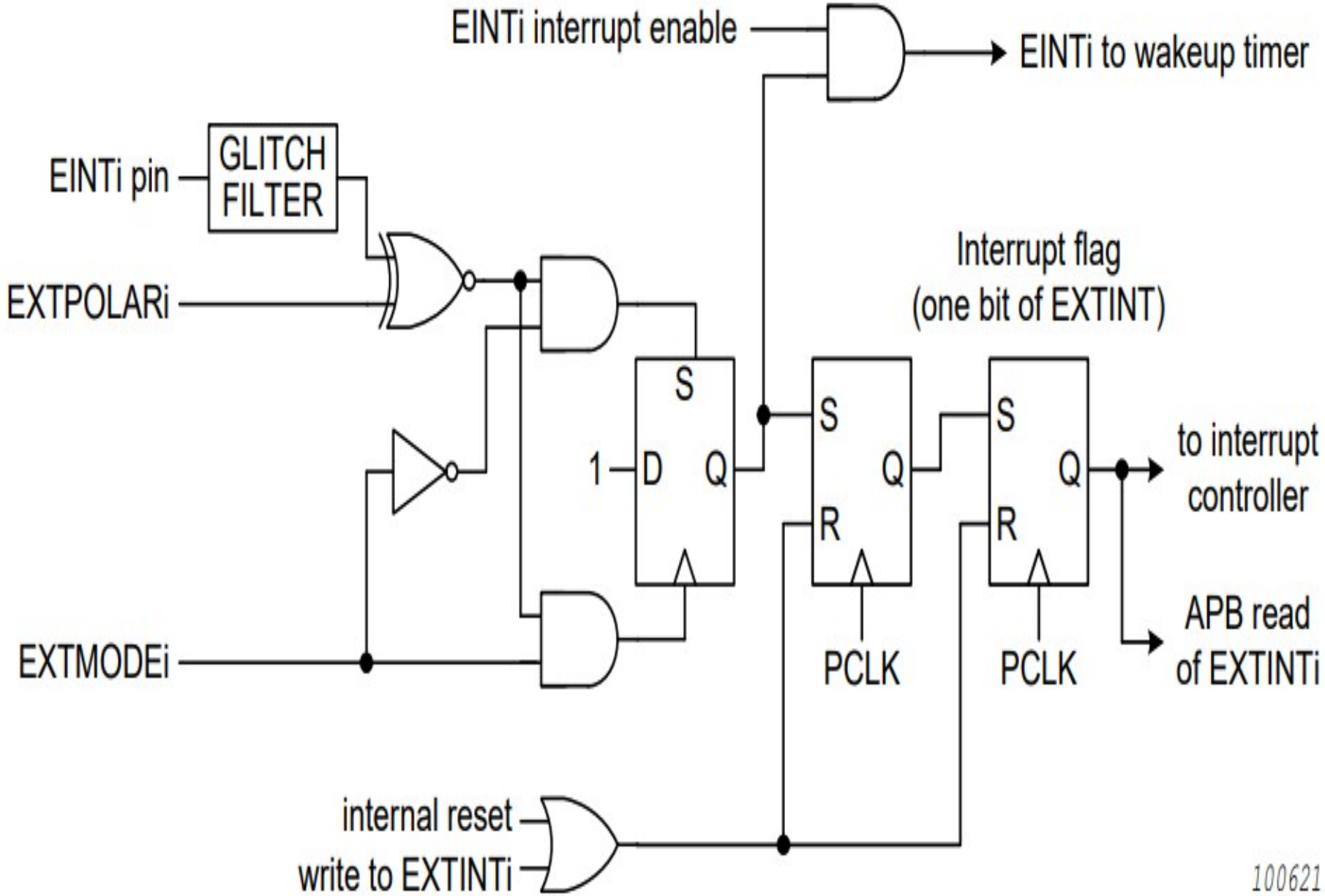
REGISTROS

Name	Description	Access	Reset value[1]	Address
EXTINT	The External Interrupt Flag Register contains interrupt flags for EINT0, EINT1, EINT2 and EINT3. See Table 10.	R/W	0x00	0x400F C140
EXTMODE	The External Interrupt Mode Register controls whether each pin is edge- or level-sensitive. See Table 11.	R/W	0x00	0x400F C148
EXTPOLAR	The External Interrupt Polarity Register controls which level or edge on each pin will cause an interrupt. See Table 12.	R/W	0x00	0x400F C14C

DIAGRAMA FUNCIONAL



External interrupt logic

gracias
merci
obrigado
danke
teşekkür ederim
sagolun
sukriya
taiku
hvala
arigatô
terima kasih
ngiyabonga
mochchakkeram
dziękuję
bedankt
go raibh maith agat
dank je
tapadh leat
takk
thank you
grazie
mauruuru
kop khun krap
dakujem
mamnun
dankon
dhanyavadagalu
merce
gracies
kia ora
murakoze
xièxie
sulpay
kam sah hamnida
dankie
diolch
akun
nandri
gracie
blagodaram
barka
dhanyavad
mesi
misaotra
mahalo
welalin
spas
rahmat
aciù
obrigada
kiitos
gratias ago
grazzi
asante
dèkuji
vinaka
enkosi
koszönöm
bayarlalaa
malondo
chnorakaloutioun
trugaraez
mersi
manana
nanni
paldies
djiere dieuf
sabodi
lack
rahmat
fadetai lawa
najis luke
ukrane
tau
tenki